

# Hydroclimatic variability drove human-megafauna-environment interactions during the late Pleistocene/Early Holocene in central Chile

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## Abstract

Major environmental changes were occurring when the first modern humans arrived in South America during the Pleistocene-Holocene transition. How these changes shaped human-environmental interactions across this period remain unclear. We analyzed the stratigraphy, biogeochemistry, and paleoclimatic models from the Ancient Tagua Tagua Lake (ATTL) in central Chile, one of the few continuous records of human and megafauna interactions with their environment in South America, to reconstruct the ATTL's ecosystem dynamics over the past 20,000 years. Results reveal that the ATTL transitioned from a shallow, cool lake with storm-driven alluvial deposition to a warmer, deeper, and more productive lake about 12,500 years ago, aligning with early humans' arrival. The ATTL became wetter but experienced severe droughts between 11,000 and 8,500 years ago, linked to shifts in Southern Westerly Winds and ENSO-like patterns. Fluctuating conditions drove humans and fauna to seek refuge in the basin, emphasizing local paleohydrology's role in shaping early human-ecosystem interactions.

Key words: Ancient Tagua Tagua Lake; Facies analysis; Biogeochemical proxies; Archaeology; Paleoclimate model; Human-ecosystem interactions

## 1. Introduction

Modern humans settled the Americas at a time of profound ecological and hydroclimatic transformations (Barnosky et al., 2004; Gutiérrez and Stockli, 2023; Mann et al., 2019). These rapid environmental changes (REC) —including the extinction and redistribution of key ecosystem-structuring species, shifts in resource availability, and alterations in atmospheric and oceanic circulation— represented one of the last major continental scale socio-ecological thresholds before modern global environmental changes (Andermann et al., 2020; Braje and Erlandson, 2013; Malhi et al., 2016; Mottl et al., 2021). Such transformations reconfigured ecological landscapes and could have shaped how early human populations adapted and expanded across South America (Baumann, 2023).

Yet, how did these REC shape early human migration, settlement, and adaptation during the initial colonization of South America? A range of diverse sources of information, including palaeoecological, archaeological, and climatic data are needed to adequately address this

question (Mayle and Iriarte, 2014). The role of climate teleconnections, including large-scale systems such as the Southeast Pacific Subtropical Anticyclone (SPSA) and the Southern Westerly Winds (SWW), have been identified as key factors influencing regional hydroclimates along the western slope of the subtropical Andes during the Last Glacial Termination (Garreaud et al., 2017; Herrera-Ossandón et al., 2023; Orellana et al., 2023; Valero-Garcés et al., 2005) (Fig. 1). Presently, prolonged positive anomalies in the SPSA block storm fronts from reaching the subtropical Andes (Fig. 1a), frequently resulting in winter drought which impacts water resources, regional livestock, and agriculture (Garreaud et al., 2017). In contrast, the SWW has entrained storm systems that travel across the southern mid-latitudes, redistributing moisture and bringing heavy rainfall that support ecosystems but also cause flooding and economic losses (Garreaud et al., 2017). These climatic drivers are modulated by phenomena such as the El Niño–Southern Oscillation (ENSO) and the Southern Annular Mode (SAM), which influence regional and global climate dynamics, including the frequency and intensity of the upper-level Pacific Subtropical Jet Stream (PSJ) and atmospheric rivers (Campos and Rondanelli, 2023; Garreaud, 2007; Garreaud et al., 2009; Gillett et al., 2021; Lu et al., 2008). In mid-latitudes, anomalies in these climatic drivers are marked by extreme events, including droughts, pluvial floods, and severe storms. Early hunter-gatherers must have been well suited and adapted to such variability (Collazo et al., 2024; Garreaud et al., 2017).

Despite considerable research and progress in identifying global paleoclimatic drivers, discerning the impact of climate on human mobility and population density in past environments remains a significant challenge (Moreno and León, 2003; Sandweiss et al., 2020). The paucity of closely related high-resolution paleoenvironmental and archaeological records hinders comparisons of whether and how climate influenced human adaptation and resource use during this early stages of exploration and colonization, leaving uncertainties about the impact of REC (Borrero, 1999; Labarca et al., 2024; Ugalde et al., 2024). Paleoclimate records from southern mid-latitudes indicate an intensification of the SWW, modulated by the effects of ENSO/SAM and the Southern Ocean (controlling CO<sub>2</sub> fluxes) during the last deglaciation (Moreno et al., 2010; Moy et al., 2002; Soteres et al., 2022; Villacís et al., 2023). Alternative hypothesis suggest that the displacement of the SWW (Kilian and Lamy, 2012; Lamy et al., 2010), was driven by shifts in the Intertropical Convergence Zone (ITCZ), in turn influenced by teleconnections with the Northern Hemisphere—particularly those associated with North Atlantic ice-rafting events and reductions in thermohaline circulation—as well as by orbital changes (Kaiser et al., 2024; Li and Liu, 2022; Schneider et al., 2014). In contrast, the interaction of the SPSA with climate dynamics that directly impact human activity—such as droughts, storms and floods

associated with storm tracks—remains poorly studied (Flores-Aqueveque et al., 2020). Past changes in the dynamics of the SPSA and SWW provide an opportunity to study the mechanisms that cause paleoenvironmental changes, and to assess the influence of mid-latitude climate variability and its interaction with human activities at an early stage (Sandweiss et al., 2020).

The Last Glacial Termination and the onset of the Holocene marked pronounced environmental shifts in South America, including early human colonization and fire use, the mass extinction of large continental mammals (>100 kg), large temperature changes (e.g. Antarctic Cold Reversal; ~14.5-12.9 cal ka BP or thousands of calibrated years before 1950), and major precipitation anomalies (e.g. Central Andes Pluvial Event, CAPE I and II, between 18-14 cal ka BP and 13-9 cal ka BP, respectively) (Latorre et al., 2006; Martin et al., 2018; Pedro et al., 2016; Prates and Perez, 2021; Ugalde et al., 2024; Villavicencio et al., 2016). Furthermore, geological and archaeological records clearly indicate that population expansion and megafaunal extinctions occurred relatively synchronously with climate variations observed during Termination I (T1) in the Northern Hemisphere, such as the Heinrich Stadial 1 (HS1, ~ 18-15 cal ka BP), Oldest Dryas (17.5-14.5 cal ka BP), Bølling-Allerød (14.7-12.9 cal ka BP), and Younger Dryas Stadial (~12.9-11.7 cal ka BP) (Alley, 2000; Barnosky et al., 2004; Cheng et al., 2020; Liu et al., 2009; Moreno-Mayar et al., 2018). However, the magnitude and distribution of these possible teleconnections at the South American mid-latitudes remain unclear and are not well constrained. Hence, the study of past environmental records, especially those linked to human activity, is of paramount importance for understanding coupled human-environment systems and the global, regional, and local mechanisms that drove ecosystem changes during the Last Glacial Termination.

Here, we examine hydroclimatic and ecosystem shifts at the ATTL geoarchaeological sites in central Chile, highlighting their importance for understanding late Pleistocene/Early Holocene human–environment interactions. ATTL is a key geoarchaeological area that preserves evidence of extinct faunal assemblages and human occupation from the Late Pleistocene to the Late Holocene (Montané, J., 1968). Our study features a comprehensive multi-proxy analysis of a newly obtained 2.8-meter stratigraphic profile from the TT19-3A sequence (Fig. 1b), offering key insights into the climatic conditions surrounding the arrival of the first hunter-gatherers in the region. This record integrates data from sedimentary facies, grain size distributions (GSDs), geochemical proxies (TOC, TIC, C/N, TS,  $\delta^{13}\text{C}_{\text{car}}$  and  $\delta^{18}\text{O}_{\text{car}}$ ), and biological indicators, including phytoliths, pollen, non-pollen palynomorphs (NPPs), and diatoms. In addition, we use an ensemble of simulations from the Paleoclimate Modeling Intercomparison Project phase 4 (PMIP4), integrated into the Coupled Model Intercomparison Project phase 6 (CMIP6), to analyze zonal wind and precipitation patterns

during the Last Glacial Maximum (LGM) and the Middle Holocene, focusing on their variability under humid and dry conditions in central Chile. This approach provides a comparative baseline of atmospheric circulation conditions before and after the arrival of the first hunter-gatherers in central Chile, thus enabling a direct contrast between these large-scale patterns and our local sedimentary record. Overall, our results highlight the interplay between hydroclimatic variability and human activity during the Last Glacial Termination, shedding deeper insight into how early populations interacted with shifting environmental conditions.

## 2. Materials and Methods

In November 2019, we recovered a sedimentary sequence measuring 15 cm in length, 10 cm in width, and 280 cm in depth from the exposed profile of the eastern wall of a 1 × 2 m excavation, corresponding to Unit 1 of the geoarchaeological test pits at TT-3 site (TT19-3A-1E, Fig. 1). Sediment samples, between 50 and 100 g of material, were obtained at 1-cm intervals along the sequence (see Supplementary Methods for more details). The uppermost 8 samples (0 - 0.38 m) were discarded due to sediment alteration due to modern agricultural practices. Sixty-six samples taken every 2 cm were analyzed for total carbon (TC), total inorganic carbon (TIC), total organic carbon (TOC = TC - TIC), total nitrogen (TN), and total sulfur (TS) using a LECO SCN furnace at the Pyrenean Institute of Ecology (IPE-CSIC) in Spain. Stable isotope compositions ( $\delta^{13}\text{C}_{\text{car}}$  and  $\delta^{18}\text{O}_{\text{car}}$ ) of the carbonate fraction from these samples were determined using a Gas Bench system interfaced with a Delta XP mass spectrometer (Thermo Finnigan, Bremen, Germany) at the Andalusian Institute of Earth Science (CSIC) in Granada, Spain. The GSDs (from 0.04  $\mu\text{m}$  to 2000  $\mu\text{m}$ ) were measured in triplicate using a Malvern Mastersizer 2000 APA2000 with Hydro 2000G AWA2000 at IPE-CSIC. Bulk sediment samples were pre-treated to remove organic matter and carbonates. Forty sediment samples were prepared for palynomorph extraction following standard chemical procedures (Heusser, 1971; Magri and Di Rita, 2015). Due to taphonomic and preservation issues, terrestrial pollen is highly underrepresented; therefore, only pollen from hygro-hydrophytes, fern spores, and other NPPs have been analyzed (Van Geel, 2001; van Geel et al., 2003).

The extraction of phytoliths and diatoms was conducted on twenty sediment samples, which were processed at the Archaeobotany and Palaeoecology Laboratory of the University of Exeter. A protocol based on standard procedures for phytolith extraction was employed, using approximately 5 g of sediment. We removed carbonates using 36% HCl and digested organic materials with 70% HNO<sub>3</sub>. Phytoliths and diatoms were separated through flotation

using ZnBr<sub>2</sub> prepared to a density of 2.3 g/cm<sup>3</sup>. We counted phytoliths, analyzing up to 200 specimens per sample, or the total number of specimens examined in cases where the slide contained fewer than 200. Finally, the phytoliths were classified according to the International Code for Phytolith Nomenclature (International Committee for Phytolith Taxonomy (ICPT), 2019; Piperno, 2006, 1988). Diatom valves on Naphrax mounted slides were observed using a Zeiss Axioscope A1 stereomicroscope with an oil immersion objective of 1000x. A minimum of 400 valves were counted when possible, identifying to species level with the exception of small fragilaroids that require higher magnification (Morales et al., 2001). We reconstructed the TT19-3A chronology using fourteenth AMS radiocarbon dates on charcoal and bulk sediments (Table S1).

## 2.1 Statistical Analysis

Radiocarbon ages were calibrated using the Southern Hemisphere calibration curve (Hogg et al., 2020), and age-at-depth assemblage was established using the Bacon and Bchron models implemented in the 'geochronR' package (Blaauw and Christen, 2011; McKay et al., 2021; Parnell et al., 2008). Robust End-member Mixing Analysis (rEMMA) was performed using the R package 'EMMAgeo' v0.9.8 to infer sediment sources, transport, and depositional processes (Dietze and Dietze, 2019; Dietze et al., 2022). Sedimentary facies were characterized based on macroscopic and microscopic descriptions (Schnurrenberger et al., 2003), and artificial facies clusters were identified using machine learning techniques (Hall, 2016). The analysis employed an unsupervised classification algorithm, utilizing geochemical data (TOC, TIC, TN, TS, GSDs, and stable isotopes) implemented in the 'scikit-learn' library (Pedregosa et al., 2011) in Python v3.9. Stratigraphic diagrams and time-series analyses were conducted and visualized in R (version 4.2.2)(see Supplementary Methods). Circulation anomalies for the Middle Holocene, and LGM periods were analyzed using PMIP4 ensembles simulations from the MPI-ESM1-2-LR, MIROC-ES2L, CESM2-WACCM-FV2, AWI-ESM1-2-LR and CESM2 models. These analyses focused on global changes in zonal wind at 500 hPa and total precipitation relative to the pre-industrial period (piControl). To define wet and dry events in central Chile (75°W-70°W, 32°S-40°S), a Regional Precipitation Index (RPI) was applied, defining wet events as RPI > 120% and dry events as RPI < 80% (see Supplementary Methods). We employed the Python libraries climlab and pyleoclim (Python 3.9) to perform wavelet and spectral analyses of past orbital parameters and paleoclimate time series. Human activities were assessed at the local scale through the analysis of lithic components found at TT-3, and at the regional scale using the p3k14c database (Bird et al., 2022) of archaeological radiocarbon dates. We calculated summed probability distributions (SPD) of archaeological radiocarbon dates using the 'rcarbon' v1.5.1

213 package (Crema and Bevan, 2021) in R 4.2.2 for South America and central Chile (between  
 214 32° and 37°S). Environmental and orbital changes from paleoclimate records and models  
 215 were summarized using available literature.

## 216 3. Results

### 217 3.1. Stratigraphy and facies: depositional variability and geochemical shifts

218

219 The TT19-3A sequence comprises 15 sedimentary facies distributed in 8 stratigraphic units,  
 220 and several sedimentary clusters (Figs. 1, 2 and supplementary material). At the base (Unit  
 221 8; 2.80 m - 2.30 m), brown to greenish sediments include clayey silty facies (L8; 2.80 m -  
 222 2.61 m) and sandy facies (L7; 2.61 m - 2.33 m), associated with clusters 4, 7, and 2 (Fig. 2).  
 223 A coarsening-upward sequence (L8b→L8a→L7) with desiccation cracks lies above these  
 224 sandy layers and may indicate subaerial exposure or subaquatic fracturing related to  
 225 degassing. Units 7 and 6 consist of greenish-brown sandy and silty sediments (facies L6;  
 226 2.33 m - 1.84 m; and L5; 1.84 m -1.65 m), linked to clusters 3 and 7 (Figs. 2 and S1). Unit 7  
 227 forms a fining-upward sequence with fine sands containing subrounded clasts and dispersed  
 228 fish vertebrae (L6b), overlain by sandy silts (L6a). Unit 6 is composed of dark grayish-green  
 229 sandy silts with reworked gastropod and bivalve shells. A marked erosive unconformity  
 230 formed at the base of Unit 5 (1.65 m - 1.50 m), which is composed of massive light grayish  
 231 green silt with millimeter to centimeter sized carbon fragments, faunal remains and lithic  
 232 artifacts and abundant *Biomphalaria taguataguensis* (*Planorbidae*) shells (L4b facies).  
 233 Facies L4 corresponds to cluster 1. The top of Unit 5 (L4a) is a dark brown, fine-grained,  
 234 carbonate-rich (6% TIC) silt layer (Figs. 2 and S1). Unit 4 (1.50 m -0.80 m) contains  
 235 carbonate (2-5% TIC) and diatom-rich coarse gray silts and fine sands, defined by four  
 236 facies. Facies L3d (1.50 m -1.30 m) consists of fine-grained light gray silts with some large  
 237 volcanic clasts up to 3.5 cm. Facies L3c (1.30 m -1.10 m) is characterized by massive to  
 238 banded gray clay silts with some millimetric clasts, higher carbonate content (up to 4% TIC),  
 239 and abundant shells of *B. taguataguensis*, randomly dispersed but more abundant towards  
 240 the top. Facies L3b (1.10 m- 0.90 m) comprises massive to banded fine-to-medium gray clay  
 241 silts with *B. taguataguensis* shells and millimeter- to centimeter-sized charcoal fragments.  
 242 Facies L3a (0.90-0.80 m) are massive medium-grained gray to dark gray sandy silts with  
 243 sub-rounded clasts, higher carbonate content and shells of *B. taguataguensis*. Units 3 and 2  
 244 contain finer gray sediments with relatively high carbonate (Unit 3, 1-5% TIC) and organic  
 245 (Unit 2, up to 2.5% TOC) contents. Unit 3 (0.80-0.45 m) is composed of gray fine sandy silts  
 246 with silty sand lenses or pockets, and shells of *B. taguataguensis* and *Diplodon* sp. (facies

L2). Unit 2 (0.45 m- 0.38 m) consists of dark gray fine sandy silts with charcoal fragments (up to 1 cm) and gastropod shells. Finally, Unit 1 (0.38 m -0 m) consists of light brownish-gray clays with abundant diatoms, higher organic content and presence of charcoal (facies L2), as well as silty sands containing rounded gravel clasts (0.5–1 cm) and larger charcoal fragments (0.2–1.5 cm; facies L1). Abundant clasts, macrocharcoal, and elevated TOC and TN in Unit 1 suggest post-depositional reworking, soil formation, and anthropogenic influence.

Overall, sediments are mostly silt (~50%) and sand (~40%), with 5–20% clay and <5% fine gravels (Fig. 2). The TT19-3A sequence displays a multimodal grain size distribution (GSD). Using rEMMA, we identified five end-members [modes; rEM1: 8  $\phi$ , rEM2: 4.1  $\phi$ , rEM3: 2.7  $\phi$ , rEM4: 1.3  $\phi$  and rEM5: 0.47  $\phi$ ] that explain ~78% of the total variance (Figs. S2 and S3). The rEM2 end-member (peak at ~60  $\mu$ m) dominates the upper half of the record (~40% variance), particularly in diatom-rich lacustrine facies (L3a, L3b, L3c). In contrast, rEM1 (peak ~4  $\mu$ m; ~18% variance) characterizes silty clay facies such as L8 and L6, while rEM3 (peak ~160  $\mu$ m; ~15% variance) occurs in fine-grained, diatom-rich intervals (1.30-0.80 m). The rEM4 end-member, with a bimodal distribution (~420  $\mu$ m and ~60  $\mu$ m), and rEM5, with a dominant coarse sand peak (~720  $\mu$ m), each explain ~14% of the variance.

Carbonate and TOC contents vary among units. Units 8 to 6 are essentially carbonate-free, with TC <0.1%. Facies L8a and L8b exhibit relatively high TS (0.1-1%) and low TC (~0.1%), alongside  $\delta^{18}\text{O}_{\text{car}}$  and  $\delta^{13}\text{C}_{\text{car}}$  values near -5‰ and -10‰, respectively. Facies L7 shows a gradual decrease in the  $\delta^{18}\text{O}_{\text{car}}$  values (cluster 2, Fig. S1), reaching the lowest values of the sequence. Unit 7 (facies L6) is characterized by two sedimentary clusters: the cluster 7, with increasing  $\delta^{18}\text{O}_{\text{car}}$  and  $\delta^{13}\text{C}_{\text{car}}$  values and cluster 3, with low rEM1 and rEM2 values and the absence of rEM5. In Unit 6 (L5 facies), rEM5 increases while  $\delta^{18}\text{O}_{\text{car}}$ , rEM1 and rEM2 values decrease. In Unit 5 (L4 facies), TC, TIC,  $\delta^{18}\text{O}_{\text{car}}$ ,  $\delta^{13}\text{C}_{\text{car}}$  and rEM2 values increase, whereas rEM1, rEM4 and rEM5 values diminish. Facies L4a records the highest TIC (>7%),  $\delta^{18}\text{O}_{\text{car}}$  (>-1 ‰),  $\delta^{13}\text{C}_{\text{car}}$  (>1‰) and rEM2 values, along with minimal rEM1, rEM3, rEM4 and rEM5. In Unit 4 (L3 facies), there is a gradual increase in the TOC/TN ratio (>5),  $\delta^{18}\text{O}_{\text{car}}$  (>-1 ‰) and  $\delta^{13}\text{C}_{\text{car}}$  values, along with variable GSD percentages. In Unit 3 (L2 facies), geochemical values show reduced variability, with high TC, TIC, TOC/TN, and rEM2 values. Finally, in Unit 2 (facies 2), TC values reach their highest levels, primarily due to the increase in TOC.

### 3.2 Aquatic–terrestrial ecosystem proxies and community shifts



282 The diatom record in the TT19-3A sequence (Figs. 3 and S4) is dominated by benthic taxa  
 283 with high specific conductance optima—such as *Anomoeoneis sculpta* (Ehrenberg) Cleve,  
 284 *Nitzschia amphibia* Grunow, *Epithemia turgida* (Ehrenberg) Kützing, *Cymbella cymbiformis*  
 285 Bahls, *Gomphonema affine* Kützing. This overall dominance is interrupted by an interval of  
 286 increased tychoplanktonic small fragilarioids between 150 and 120 cm (L3d, L3c). At the  
 287 base of the sequence (Unit 8, L8b), diatom valves are absent. Moving upward (L8a, Units 7  
 288 and 6), we observe the presence of *A. sculpta* (most abundant), *N. amphibia*, *E. turgida*, *G.*  
 289 *affine*, and *Ulnaria* sp. In Unit 5 (L4b), *A. sculpta* remains the dominant taxon (~84%  
 290 abundance), and diatom valves are ubiquitous, allowing for the estimation of relative  
 291 abundances. A notable assemblage shift occurs in L4a (Unit 5), L3d, and L3c (Unit 4), where  
 292 *A. sculpta* nearly disappears and diatom diversity increases substantially. Here, small  
 293 fragilarioids, *N. amphibia*, *E. turgida*, *C. cymbiformis*, and *G. affine* all become more  
 294 common, accompanied by planktonic species such as *Stephanocyclus meneghinianus*  
 295 (Kütz) Kulikovskiy, Genkal and Kociolek, *Aulacoseira granulata* (Ehrenberg) Simonsen, and  
 296 *Ulnaria* sp.). L4a and L3d differ from L3c in that small fragilarioids and *A. granulata* are more  
 297 abundant, whereas *N. amphibia*, *S. meneghinianus*, and *A. sculpta* (~5% abundance) are  
 298 more common in L3c. Subsequently, L3b records a marked recovery of *A. sculpta* (~77%),  
 299 accompanied by a decrease in other taxa. In L3a, *A. sculpta* declines (~22%) as *G. affine*  
 300 and *N. amphibia* increase. In Unit 3 (L2) and Unit 2 (L1), *A. sculpta* partially recovers  
 301 (~36%) alongside slight increases in planktonic taxa *S. meneghinianus* and *A. granulata*.  
 302 The pollen and NPP data (Fig. 3, S5) reveal distinct assemblages of hygro-hydrophytes,  
 303 ferns, freshwater algae, and spores of coprophilous fungi. Two well-defined sections are  
 304 apparent. The first section (263-150 cm; Units 8, 7, 6, and 5) is characterized by high  
 305 abundances of freshwater algae *Botryococcus* and *Spirogyra*, along with occasional  
 306 *Mougeotia* and *Pediastrum*. Units 8 and 7 (263–185 cm), *Typha*, ferns (*Pteris*),  
 307 *Botryococcus*, and *Spirogyra* dominate. Towards the upper part of this first section (238-150  
 308 cm), abundance of *Botryococcus*, *Spirogyra*, and arbuscular mycorrhizal fungi (*Glomus*)  
 309 remains elevated, but littoral and floating vegetation reach their lowest values. The second  
 310 section (150-38 cm) shows a progressive increase in water surface indicators and the  
 311 colonization of littoral (Cyperaceae, Juncaceae, *Typha*) and floating vegetation (*Lemna*,  
 312 *Nuphar*) especially pronounced between 55 and 38 cm. Freshwater algae remain present  
 313 throughout this interval.

314 Phytolith analysis (Figs. 3 and S6) identifies opal phytoliths from Cyperaceae, Poaceae, and  
 315 Arecaceae. Within Poaceae, subfamilies Bambusoideae, Chloridoideae, and Pooideae were  
 316 distinguished. At the species level, we detected *Jubaea chilensis* within Arecaceae, as well  
 317 as phytoliths from woody dicotyledons. Unit 8 exhibits very low phytolith concentrations

compared to the rest of the record. Here, Poaceae phytoliths dominate (~90%), with fewer Bambusoideae and Chloridoideae phytoliths (~5%). Unit 7 (230-175 cm), Bambusoideae and Chloridoideae reach their highest percentages (46% and 26%, respectively), while Cyperaceae increases modestly (~13%). Unit 6 (175–165 cm) shows the highest overall Poaceae content, though Bambusoideae, Pooideae, and Chloridoid percentages decrease. From Unit 5 to Unit 3 (165–40 cm), the total phytolith content rises, accompanied by the first appearance of woody and shrub morphotypes indicative of sclerophyll forests (woody dicotyledons and *J. chilensis*). During this interval, Bambusoideae and Chloridoid proportions slightly decline in Unit 3. Additionally, Cyperaceae levels increase again (to ~13%) between Units 5 and 4 (150–142 cm) before stabilizing at lower values (~2%) thereafter.

### 3.3. Chronological context

The chronological framework for the TT19-3A sequence provided by the age-depth models (Bacon and Bchron; Fig. 4) supports these interpretations, offering a robust temporal backdrop for correlating cultural and environmental shifts. The models excluded two outlier ages. Bulk sediment and charcoal samples taken at the same depths showed no significant differences, ruling out a reservoir effect. According to the ensemble age-depth model, the TT19-3A sequence covers the last 20.5 cal ka BP (Table S1 and Fig. 4), with a higher model robustness in the upper four units. Unit 8 (2.66 m - 2.30 m) spans from ~20.5 to 16 cal ka BP with an accumulation rate of ~200 a cm<sup>-1</sup>. Unit 7 (2.30 m - 1.84 m) was deposited between ~16 and 12.5 cal ka BP with an accumulation rate of ~100 a cm<sup>-1</sup>. Unit 6 (1.84 m - 1.65 m) was deposited between 13 and 12.6 cal ka BP with an accumulation rate of ~20 a cm<sup>-1</sup>. Unit 5 (1.65 m - 1.50 m), encompassing the Pleistocene-Holocene transition (12.6 - 11.3 cal ka BP), has an accumulation rate of ~100 a cm<sup>-1</sup>. Unit 4 (1.50 m - 0.80 m) extends from 11.3 to 8.6 cal ka BP, characterized by ~50 a cm<sup>-1</sup>. Unit 3 (0.80 m - 0.45 m) spans between 8.6 and 6.5 cal ka BP with a median rate of ~60 a cm<sup>-1</sup>. Unit 2 (0.45 - 0.38 m) covers between 6.5 and 6.0 cal ka BP with an accumulation rate of ~80 a cm<sup>-1</sup>.

## 4. Discussion

### 4.1. Depositional dynamics and associated hydroclimatic changes during the Last Glacial Termination (20.5–12.8 cal ka BP)

The ATTIL record highlights the climatic shifts at mid-latitude South America during the Last Glacial Termination (T1; ca. 20.5 - 12.5 cal ka BP), showing shallow littoral lacustrine conditions and high-energy sediment input prior to the first human settlements in central

Chile (Figs. 5 and 7). Drier conditions prevailed between 20.5 and 18.5 cal ka BP as indicated by fine-grained deposition consistent with greater meltwater drainage and dust input (Amann et al., 2022; Sun et al., 2002; Vandenberghe, 2013; Vandenberghe et al., 2018), whereas low  $\delta^{13}\text{C}_{\text{car}}$  and  $\delta^{18}\text{O}_{\text{car}}$  values reflect reduced evaporation and increased freshwater input from snowmelt (Dansgaard, 1964; Gonfiantini et al., 2001; Rozanski et al., 1993) (Fig. 5F-H). During the final stage of the LGM, the ATTIL basin was characterized by the deposition of fine L8 facies, with high silt and clay contents (>80%), very low TC (<0.1%), and relatively elevated TS percentages (Figs. 2, 3, and 4). The unimodal grain size distribution (~4  $\mu\text{m}$  mode) and low sorting suggest gradual, year-round sediment input, either via long-distance aeolian transport (Zielhofer et al., 2017) or meltwater flows from glaciated areas (Amann et al., 2022). As in other glaciated settings, the fine-grained material deposited during glacial times was likely related to glacier dynamics, aeolian dust (Amann et al., 2022; Sun et al., 2002; Vandenberghe et al., 2018), and local clays derived from weathered soils in the catchment (Sun et al., 2002). Stratigraphic correlations show that the L8 facies can be correlated with “*Member 5*” (Varela, 1976) and “*Unit 5*” (Valero-Garcés et al., 2005) (Fig. 1C) from previously described ATTIL profiles, dated to ca.  $21.5 \pm 0.65$  and  $20.4 \pm 1.3$  cal ka BP, respectively (Heusser, 1983; Montané, 1969; Valero-Garcés et al., 2005). Researchers originally interpreted these deposits as a massive clay layer rich in *Nothofagus dombeyi/obliqua* type pollen, suggesting cold and wet conditions or shallow lacustrine environments. However, the limited presence of planktonic diatoms, aquatic pollen, and phytoliths, along with only minor benthic diatoms and freshwater algae, points instead to a shallow littoral lake setting under persistent, fine detrital input influenced by a glaciated watershed. A brief high-energy alluvial event at ~19.5 cal ka BP did not significantly elevate lake levels or bioproductivity (Figs. 3 and 5F-H). Varela (Varela, 1976) describes subvertical cracks filled with authigenic magnesium oxide and iron minerals in *Member 5*, consistent with those in the L8 and L7 facies (Unit 8). Subvertical cracks filled with authigenic minerals can be interpreted as desiccation cracks, sub-aquatic sedimentary fractures, or fragipan features (Beal et al., 2009; Ciolkosz et al., 1990; Fathi-Moghadam et al., 2011; Wang et al., 2022). These cracks, lacking mud curls and filled with magnesium oxide, iron, and sulfur minerals, suggest salinity fluctuations or anoxic conditions promoted shrinkage through deflocculation or intracrystalline dewatering (Beal et al., 2009; Wang et al., 2022). The high sulfur content of these facies suggests methane emissions and anaerobic sulfate or metal reduction by methanotrophic archaea (Wang et al., 2022), reinforcing the notion of a littoral depositional environment strongly shaped by watershed inputs.

386 A high-energy depositional phase occurred between 18.5 and 16.5 cal ka BP, marked by  
 387 sandy L7 facies (about 70% - 80% sand) with low TC values, and the lowest  $\delta^{13}\text{C}_{\text{car}}$  and  
 388  $\delta^{18}\text{O}_{\text{car}}$  values in the record (Figs. 2 and 4). The coarse, bimodal grain size distribution (rEM4;  
 389 main peak  $\sim 420\ \mu\text{m}$ , secondary  $\sim 60\ \mu\text{m}$ , see Fig. S3) suggests heavy rainfall/intense storm,  
 390 or rapid melt-driven discharge that delivered coarse sediments into the lake's littoral zone  
 391 (Dietze et al., 2014; Sun et al., 2002; Vandenberghe et al., 2018). The lowest  $\delta^{18}\text{O}$  values  
 392 likely reflect colder waters, an increase in the amount effect or a different moisture source  
 393 (Dansgaard, 1964), whereas lower  $\delta^{13}\text{C}$  values possibly reflect increased  $\text{CO}_2$  input from  
 394 terrestrial respiration or lower aquatic productivity. Later, from 16.5 to 14.5 cal ka BP, a shift  
 395 to coarser sediments and stormier conditions occurred again without increasing lake levels  
 396 (Figs. 5A, B, E). This interval corresponds to a period of intense hydrological and  
 397 depositional variability, linking local sedimentary patterns to the broader regional climate  
 398 context.

399 To understand the atmospheric patterns during this period, we analyzed precipitation and  
 400 zonal wind outputs from the PMIP4-CMIP6 multimodel ensemble simulations for the LGM  
 401 (Figs. 6, 7S–9S; Table S2). The multimodel ensemble indicates that during the LGM, dry  
 402 conditions were associated with intensified zonal wind speeds at low and mid-latitudes,  
 403 accompanied by negative anomalies at subtropical and high latitudes (Figs. 6a, 6c). In  
 404 contrast, wet anomalies are linked to weakened tropical trade winds, akin to warm ENSO-  
 405 like conditions, accompanied by positive anomalies along extratropical storm tracks (Figs.  
 406 6E, 6G). Zonal wind speeds at tropical and subtropical latitudes show positive anomalies  
 407 over the southern Pacific, while negative anomalies prevail over the central-southern Andes.  
 408 A similar atmospheric pattern to those of the glacial period likely contributed to extreme  
 409 events between 18.5 and 14.5 cal ka BP (Figs. 5A, 5B). Sedimentological and limnological  
 410 evidence, however, suggests this was insufficient to maintain elevated lake levels.

411 Between 16.5 and 12.8 cal ka BP, sedimentation reverted to sandy silts (Unit 7, L6a and L6b  
 412 facies), with varying clay percentages and low TS and TC values. Microfossils remained with  
 413 low abundance, except for two samples showing a temporary increase in Poaceae  
 414 phytoliths, benthic diatoms, and Cyperaceae, hinting at short-lived stabilization of littoral and  
 415 riparian communities (Figs. 3 and S4). The presence of freshwater algae (*Botryococcus* and  
 416 *Spirogyra*), littoral vegetation such as *Typha*, and sporadic fern spores (*Pteris*) indicates  
 417 wetlands and riparian areas adjacent to the basin (Fig. S5).  $\delta^{13}\text{C}_{\text{car}}$  and  $\delta^{18}\text{O}_{\text{car}}$  values rose  
 418 relative to Unit 8, indicating changes in hydrological balance (precipitation - evaporation  
 419 balance), seasonality or source area. From 16.5 to 14.5 cal ka BP (L6b facies), grain size  
 420 distributions (rEM5) were bimodal ( $\sim 700\ \mu\text{m}$  and  $\sim 65\ \mu\text{m}$ ), more consistent with fluvial or

reworked fluvial/lacustrine-aeolian processes (Amann et al., 2022; Dietze et al., 2014; Sun et al., 2002; Vandenberghe, 2013; Vandenberghe et al., 2018). Such distributions occur where braided rivers and fluvial processes overlap, mixing coarse and fine sediments (Sun et al., 2002; Zhang et al., 2018). Proximal aeolian coarse-grained silt transport typically shows a unimodal grain-size distribution between 40 and 90  $\mu\text{m}$  (Sun et al., 2002; Zhang et al., 2018). In contrast, GSDs exceeding a  $\sim 250 \mu\text{m}$  mode suggest high-energy fluvial processes. Moreover, in other environments and depositional process simulations, grain sizes ranging from 250 to 850  $\mu\text{m}$  have been linked to intense runoff, alluvial fan formation, and are also observed in sandy coastal settings as well as wind-eroded sediments (Orton and Reading, 1993; van der Vegt et al., 2020; van IJendoorn et al., 2023). After 14.5 cal ka BP (L6a facies), a shift toward finer particles (rEM1,  $\sim 3.8 \mu\text{m}$ ; rEM2,  $\sim 60 \mu\text{m}$ ) suggests improved hydrodynamic sorting, potentially linked to rising lake levels or greater availability of finer sediments. These conditions reflect a dynamic interplay between wind-driven, glacial-mediated and fluvial processes influencing deposition at the ATTL site during glacial time. Further analysis of the quartz surface texture is necessary to differentiate between aeolian and fluvial processes.

Our interpretation contrasts with earlier suggestions of cold and wet conditions at the end of the LGM (Heusser, 1983, 1971; Valero-Garcés et al., 2005; Varela, 1976). Heusser (1983) proposed the presence of a closed-canopy *Nothofagus* forest between 28 and 11 ka BP, suggesting cool and wet conditions potentially driven by a northward shift of the SWW. Recalibrated pollen chronologies, however, suggest that the percentages of the *Nothofagus antarctica/dombeyi* type declined earlier than previously reported (Fig. S6), occurring between approximately 19-16 cal ka BP (initially dated at  $14.5 \pm 0.35$  cal ka BP; Unit 2a, Figs. 7H and S6) in Heusser (1983) and between 14.9-12.7 cal ka BP (initially dated at  $11.71 \pm 0.43$  cal ka BP; Unit 3, Fig. 1) in Valero-Garcés et al. (2005). In the central Chile valley, *Nothofagus dombeyi/obliqua* exhibits a more restricted distribution compared to other members of the genus and subgenus *Lophozonia* (Mathiasen et al., 2020). Germination studies reveal that these species are not strictly dependent on high moisture levels, challenging the assumption that their expansion was moisture-driven (León-Lobos and Ellis, 2005; Reyes-Díaz et al., 2005). León-Lobos and Ellis (2007) found that the germination of *Nothofagus* species (*N. alpina*, *N. dombeyi*, *N. leonii*, *N. glauca*, and *N. obliqua*) varies more significantly with temperature, suggesting that moisture was important but not critical for their expansion into central Chile.

Intensified alluvial activity, associated with more frequent and severe storms at ATTL, is practically coeval with the HS1 (Broecker, 1995; Martin et al., 2018), CAPE I (Gayo et al.,

2012; González-Pinilla et al., 2021; Quade et al., 2008; Ugalde et al., 2024) and ACR/Bölling–Alleröd events (Pedro et al., 2016) (Fig. 7), during a period characterized by increasing winter insolation and decreasing summer insolation (Figs. 7D, 7E, 7O, and 7P). These changes have been linked to shifts in global temperature gradients driven by residual Milankovitch effects (precession and obliquity; Figs. 7A and 7B). Some authors have posited that these shifts may have triggered a southward migration of the ITCZ and SWW, a weakening of the American Monsoon System, and modifications to the PSJ (Kaiser et al., 2024, 2008; Lamy et al., 1999; Novello et al., 2017). This atmospheric configuration, coupled with the reactivation of ENSO-like conditions ca. 16 cal ka BP (Figs. 7J and 7Q), likely amplified the frequency and intensity of alluvial events, displaying atmospheric patterns similar to those observed in simulations for the LGM (Fig 6G). Marine sediment cores off the Chilean coast date T1 to between 19.2 and 18.5 cal ka BP, indicating arid conditions (Lamy et al., 1999). Similarly, Andean glaciers (33°S - 36°S) retreated between 28.0 and 18.0 ka BP (Fig. 7I) without subsequent re-advances during T1 (Fernández-Navarro et al., 2024; Herrera-Ossandón et al., 2023; Singer et al., 2000). Changes in leaf wax  $\delta D$  and  $\delta^{13}C$  isotopes from marine sediment cores suggest hydrological changes influenced by precessional cycles, with SWW retreating during T1 (Figs. 7N and 7G), reaching a minimum at 11 cal ka BP and again between 5-6 cal ka BP (Kaiser et al., 2024). Marine records also show a 6°C–7°C rise in SST between 19 and 12.5 cal ka BP, interrupted by cooling around 14.8 - 13.3 cal ka BP, corresponding to the ACR (Kaiser et al., 2024, 2008, 2005; Kim et al., 2002). The decline in high-energy processes after ca. 14.8 cal ka BP was accompanied by fluvial and fine-grained sediment deposition (Fig. 7J-K-L). Orbital changes alone cannot fully explain these phenomena. Instead, multiple high-magnitude events associated with rapid coupled oceanic–atmospheric changes likely interacted with gradual orbital processes to produce the alluvial and fluvial deposits in TT19-3A between ~18.5 and 14.5 cal ka BP. Further studies are required to better understand the interplay among these factors and their implications for regional paleoecology.

#### 4.2. Changing lake environments, new habitats and first human settlement (12.8–6.5 cal ka BP)

Archaeological evidence from the ATTL indicates that human occupation at sites TT-1 (Montané, 1968; Varela, 1976), TT-2 (Nuñez et al., 1994), and TT-3 (Labarca et al., 2023) began around 12.6 - 11.6 ka cal BP during deposition of Unit 5 in the TT19-3A sequence (Fig. 1c). Between ca. 12.8 and 12.4 cal ka BP (Unit 5), coarser L5b and L5a facies (with a grain size mode at ~700  $\mu m$ ; rEM5) indicate high-energy conditions and stronger alluvial influence in the lake shoreline (Figs. 2, 4, and 5). The increasing ubiquity of the diatom

species *Anomoeoneis sculpta* (Figs. 3 and S4) —known for inhabiting alkaline (1350 mg/L CaCO<sub>3</sub>), shallow freshwater environments with elevated pH (9.3) and conductivity (5760 µmS/cm) (Fritz, 2013; Stenger-Kovács et al., 2023)— and the rise of planktonic diatoms (e.g., small fragilarioids) suggest a permanent, alkaline lake established in the basin (Fig. S4). The presence of *Sporormiella* spores, megaherbivore remains, and evidence of early human occupation around 12.5 cal ka BP (Labarca et al., 2024) coincides with this transitory alkaline lake phase. From 12.5 to 11.2 cal ka BP (Unit 5), a significant depositional and biogeochemical shift occurred. Greenish to gray sandy silts with coarser GSDs (rEM<sub>2</sub>; L4b facies) had abundant gastropods and charcoal spicules, higher carbonate (TIC) content, more positive  $\delta^{13}\text{C}_{\text{car}}$  and  $\delta^{18}\text{O}_{\text{car}}$ , increased diatom abundance (especially *A. sculpta*), and a rise in woody, shrub, and *Jubaea chilensis* pytholiths (Figs. 2, 3, and 4). This suggests a moderately alkaline, shallow lake environment characterized by higher bioproductivity, more stable water levels, and warmer, more humid conditions. In general, freshwater lakes with high conductivity and salinity levels exhibit a negative correlation with diatom diversity (Fritz, 2013). The presence of *Jubaea chilensis* corresponds to the endemic Chilean palm, currently distributed between ~30°S and 35°S (Jara-Arancio et al., 2022). Their occurrence near the 1.65 m depth (ca. 12.4 cal ka BP; see Fig. 4) may indicate increased humidity, reduced megafaunal pressure, or human transport for food, temporary habitation, or other uses (Robinson et al., 2021; Trepel et al., 2024). Alternatively, other taphonomic and sedimentary transport processes associated with a rising lake level could also explain its presence. A brief interval between ~11.4 and 11.2 cal ka BP (L4a facies) shows even higher TIC,  $\delta^{13}\text{C}_{\text{car}}$ ,  $\delta^{18}\text{O}_{\text{car}}$ , and rEM<sub>2</sub> values, as well as a shift in diatom assemblages toward tychoplanktonic small fragilarioids and *A. granulata*. These changes suggest a decrease in salinity and an increase in lake turbulence (Figs. 2, 3, and 4). During this period, freshwater algae dominate the palynomorph assemblage, and woody and shrub taxa appear alongside Poaceae, Bambusoideae, and Chloridoid phytoliths (Fig. 3).

Isotope signatures support warmer conditions and higher lake levels at the onset of the Holocene. In ATTL sediments, calcite is the principal carbonate phase. Elevated  $\delta^{13}\text{C}_{\text{car}}$  (~1‰) and  $\delta^{18}\text{O}_{\text{car}}$  (~-0.2‰) values in authigenic calcite indicate evaporation-driven isotopic enrichment of dissolved inorganic carbon (DIC). This enrichment, facilitated by isotopic equilibrium exchange with CO<sub>2</sub>, is regulated by temperature, the isotopic composition of inflows, pH, and lacustrine bioproductivity (Horton et al., 2016; McCormack et al., 2019; McCrea, 1950; Valero-Garcés et al., 2005). Currently, there is no data to validate the isotopic enrichment model for meteoric precipitation proposed by Valero-Garcés et al. (2005) for ATTL. However, McCormack et al. (2019) suggest that  $\delta^{18}\text{O}_{\text{car}}$  and  $\delta^{13}\text{C}_{\text{car}}$  values are closer to freshwater runoff compared to other carbonates like aragonite or dolomite. The

isoscape from Chilean lakes serves as an analogue for time-averaged local precipitation isotopes where direct data are lacking (Scott et al., 2021). In central Chile, meteoric water  $\delta^{18}\text{O}$  averages from -4‰ to -2‰ in summer and -7.6‰ to -5.9‰ in winter (Scott et al., 2021; Terzer-Wassmuth et al., 2021). Data from the *HOLOCHILL project* show notable isotopic variability in lakes by altitude (Fig. 10S). Coastal areas, such as Lago Vichuquén, exhibit  $\delta^{18}\text{O}$  values ranging from -2‰ to 0‰, while Laguna Matanzas shows values between 7‰ to 8‰. In contrast, mountain lakes like Laguna del Maule and Laguna Lo Encañado have  $\delta^{18}\text{O}$  values of -11‰ to -12‰. Altogether, these data suggest a rise in lake levels under warm and humid conditions at the onset of the Holocene, consistent with an environment transitioning toward a new ecological state (Labarca et al., 2024).

No archaeological radiocarbon dates have been identified between ca. 11.5 and 9.0 ka cal BP at the ATTL; however, open-air residential camps and extensive lithic knapping areas appear at ca. 9.0 ka cal BP (Labarca et al., 2024). Between 11.2 and 8.6 cal ka BP (Unit 4), a series of fluctuations in water levels, detrital input, and carbonate isotopic signals are documented by several proxies (Figs. 2 and 3). Facies L3a-d show alternating clayey silty sands and sandy silts with variable TIC,  $\delta^{13}\text{C}_{\text{car}}$  and  $\delta^{18}\text{O}_{\text{car}}$  values, as well as shifts in diatom communities and phytolith assemblages (Fig. 3). L3d facies (ca. 11.2 and 10.4 cal ka BP) show a shift in the isotopic signal of authigenic calcite, accompanied by an abrupt increase in rEM<sub>5</sub> percentages. Episodes of high-energy alluvial transport, variations in the presence of *A. sculpta* and small fragilarioids, and frequent planktonic diatoms reflect changing littoral environments and variable lake levels (Figs. 3 and S4). Small fragilarioids —facultative planktonic organisms from lentic and deltaic environments— and *A. granulata* indicate a rise in lake levels, reduced light availability, and increased wind or tidal turbulence, consistent with high rEM<sub>5</sub> values (Benito et al., 2015; Starratt et al., 2021). Woody and shrub phytoliths, along with Poaceae and Bambusoideae, reflect nearby vegetation responses to these hydrological and climatic oscillations (Figs. 3). The L3c facies (10.4 and 9.8 cal ka BP) show variable percentages of rEM<sub>2</sub> and rEM<sub>3</sub>, accompanied by elevated rEM<sub>4</sub> values, which highlight high variability of alternating sedimentary processes (littoral, fluvial, aeolian). During deposition of facies L3b (between ca. 9.8 and 9.0 cal ka BP), changes in the fluvial sediment delivery (rEM<sub>3</sub>) suggest some hydroclimatic instability. *A. sculpta* increases rapidly in this facies, while small fragilarioids practically disappear, indicating a shallow to ephemeral and subaerial depositional environment. The phytolith assemblage remains virtually unchanged. Finally, deposition of L3a facies, (9.0 - 8.6 cal ka BP) characterized by a high content of fine sand (rEM<sub>4</sub>), and a decrease in planktonic diatom species. These changes culminated in the Early Holocene, where stable lake levels, finer sediments (rEM<sub>2</sub> and a minority of rEM<sub>1</sub>), a dominance of hygro-hydrophilous communities and elevated organic productivity (Units 3



and 2, ca. 8.6 - 6.0 cal ka BP) fostered greater diatom diversity and more mesotrophic, alkaline conditions. Finer sedimentation and sustained presence of planktonic diatoms (i.e., *Stephanodiscus meneghinianus*) suggest relatively higher lake levels compared to Unit 4, and less sediment input from the watershed. At the time of deposition of Unit 3 (8.6 - 6.5 cal ka BP), the later occupation of the archaeological site TT-3 occurred.

#### 4.3 Early human–ecosystem interactions in ATTL during the Pleistocene–Holocene transition

The progressive transformation of the landscape toward the end of the Pleistocene and the onset of the Holocene —evidenced by rising lake levels and the diversification of available resources (Labarca et al., 2024)— represented a turning point for human occupation in the ATTL. These changes broadened subsistence possibilities for the earliest hunter-gatherers, marking the onset of a distinct ecological stage in South America. In the ATTL, the transition from the Pleistocene to the Holocene (14.5 to 11.7 cal ka BP) is characterized by a gradual rise in lake levels (Figs. 5D and 7L). An alluvial event between ca. 12.8 and 12.5 cal ka BP and the development of beach-like depositional environments in littoral areas indicate a return to drier conditions (Figs. 5 and 7). The end of this event coincided with the discovery of a combustion feature along with megafaunal remains, dated to 12.5 - 12.4 cal ka BP (Labarca et al., 2024).

In a broader regional context, the SPD of archaeological radiocarbon dates for South America and central Chile (32°S - 37°S), reveal a pronounced increase around 12.6 ka cal BP (Figs. 7F). This uptick aligns with the timing of initial human presence documented at the ATTL and the environmental shifts described earlier, underscoring a notable rise in human activity as climates warmed and landscapes transformed. Archaeological evidence indicates that human occupation of the ATTL began during the Fishtail point horizon (Prates et al., 2020). The archaeological sites in the ATTL (TT-1, TT-2, and TT-3) dated to ca. 12.6 - 11.6 cal ka BP, provide unequivocal evidence of direct human–megafauna interactions, including gomphotheres, horses, and cervids (Labarca et al., 2024; Lizama-Catalán and Labarca, 2023; Méndez and Jackson, 2015; Montané, 1968; Núñez et al., 1994). These sites also document the consistent exploitation of small game (primarily birds and frogs) and the use of local plant resources (Labarca et al., 2024; Lizama-Catalán and Labarca, 2023; Méndez and Jackson, 2015; Montané, 1968; Núñez et al., 1994). Interpreted as logistic or ephemeral camps, these settlements were part of a broader and more complex mobility system that included at least the Andean range (Labarca et al., 2024; Méndez and Jackson, 2015). Regionally, however, late Pleistocene settlements are conspicuously absent, whereas Early

597 Holocene archaeological sites are common in the Andes, along the coastal range, and the  
 598 coastline (Barnosky et al., 2004; Moreno-Mayar et al., 2018).

599 Amid these regional cultural developments, ATTL paleoclimatic proxies record increased  
 600 water availability and warming during between ca. 15 and 11 cal ka BP (Fig. 7L), followed by  
 601 a marked increase in extreme hydroclimatic events between 11 and 9 cal ka BP (Figs. 7J-K).  
 602 Marine records off the northern Chilean coast show a progressive decline in SWW influence  
 603 near 30° S toward the end of the Late Pleistocene, as evidenced by elevated  $\delta D$  values of  
 604 the n-C31 alkane from GeoB7139-2 marine sediments between 14 and 12 cal ka BP (Kaiser  
 605 et al., 2024; Figs. 2, 3, 7L–N). During the Early Holocene, Kaiser et al. (2024) document two  
 606 concurrent climate shifts: a rapid intensification of the SWW at 45°S and a progressive  
 607 northward expansion of these winds to 30° S between 12 and 6 cal ka BP —likely driven by  
 608 a weakened subtropical anticyclone. Furthermore, both shifts coincided with large-amplitude  
 609 REC episodes linked to the Younger Dryas and ENSO-like variability (Cheng et al., 2020;  
 610 Kaiser et al., 2024; Rein et al., 2005; Wang et al., 2020). These processes—partially driven  
 611 by obliquity-induced orbital changes and ENSO-like variability—likely governed the variability  
 612 of both the SWW and the SPSA in central Chile between ca. 12.5 and 9 cal ka BP, forming a  
 613 complex interplay not fully captured by other records at this latitude (Fernández-Navarro et  
 614 al., 2024; Herrera-Ossandón et al., 2023; Kaiser et al., 2024).

615 Against this dynamic environmental backdrop, the earliest human groups present at ATTL  
 616 adopted logistical settlement patterns under increasingly favorable conditions, taking  
 617 advantage of a wide range of lacustrine resources. This strategy persisted until around 11-9  
 618 cal ka BP (Labarca et al., 2024; Núñez et al., 1994). The intertwined development of climate,  
 619 megafauna, and human strategies underscores the profound impact that shifting  
 620 environmental regimes had on the cultural trajectories of early hunter-gatherer communities.

#### 621 4.4 Early-Middle Holocene: increased environmental variability and emerging human 622 settlement patterns

623 Early and Middle Holocene environmental reconstructions in central Chile suggest a  
 624 dynamic interplay among regional teleconnections, climate-driven erosive processes,  
 625 tectonics, volcanism, and the evolving impacts of early hunter-gatherers on the ecosystem  
 626 (Varela, 1976; Heusser, 1983; Jenny et al., 2002; Valero-Garcés et al., 2005; Maldonado et  
 627 al., 2010; Ortega et al., 2012; Frugone-Álvarez et al., 2020). The TT19-3A sequence shows  
 628 that, compared to the Late Pleistocene, the Early Holocene was characterized by higher lake  
 629 levels, intensified evaporative regimes, and the emergence of more diverse ecological  
 630 assemblages (Figs. 5 and 7). The higher abundance of woody dicotyledon phytoliths

631 indicates the prevalence of sclerophyll forests, whereas the occurrence of *J. chilensis*  
 632 highlights the abundance of palm trees, likely more prevalent due to the absence of  
 633 megaherbivores, humid climatic conditions, and/or human use (e.g., for food or other  
 634 purposes) (Robinson et al., 2021). Phytoliths from Poaceae and Pooideae groups indicate  
 635 the coexistence of grassland and sclerophyll shrubs. An increase in Bambusoideae  
 636 morphotypes suggests the expansion of *Chusquea* sp., while the presence of Chloridoid  
 637 phytoliths—commonly found in arid or semi-arid regions—points to possible drought periods.  
 638 Throughout the Early to Middle Holocene (ca. 11.7-6.0 ka cal BP), the SPD remains  
 639 elevated, albeit with intermittent variability. When considered alongside local archaeological  
 640 evidence, these trends imply a continuous human presence with episodic shifts in  
 641 occupation intensity in central Chile. Between 11.5 and 11 cal ka BP, the proportion of fluvial  
 642 deposits reached its highest levels (Figs. 5D and 7L), possibly triggered by a northward shift  
 643 of the SWW (Kaiser et al., 2024) and weakening of the SPSA (Figs. 7N and 7G). After this  
 644 peak fluvial input, the record indicates shifts in high-energy sedimentary transport and an  
 645 increase in diatoms (small fragilarioids) associated with turbulent lacustrine environments  
 646 between 11 and 8.5 cal ka BP (Figs. 7J-L). It is possible that high ENSO-like activity along  
 647 the Peruvian coast (Fig. 7R) aligns with the changes recorded in the ATTL and drought  
 648 periods at Laguna Aculeo (Jenny et al., 2002; Rein et al., 2005). Such large-scale events  
 649 might have triggered a scaling break effect, potentially strengthening STJ conditions and  
 650 contributing to drought events (Collazo et al., 2024; Zhu et al., 2019). Variability in STJ  
 651 intensity, influenced by ENSO-like phenomena and possibly combined with residual effects  
 652 of glacial-interglacial orbital changes (Milankovitch cycles), may have contributed to the  
 653 formation of alluvial and fluvial deposits observed during this period in central Chile (Gillett et  
 654 al., 2021; Zhu et al., 2019).

655 During the first half of the Middle Holocene (8.5 – 6 cal ka BP), central Chile gradually  
 656 became more arid. This trend, however, was less severe than the drying episode between  
 657 11.0 and 8.5 ka cal BP (Figs. 5, 7H-J). This is supported by PMIP4-CMIP6 multi-model  
 658 ensemble outputs (Figs. 6, 11S-13S; Table S2), show that, during simulated wetter intervals,  
 659 zonal winds strengthened slightly relative to the pre-industrial control (Fig. 6F). At lower  
 660 latitudes, negative zonal wind anomalies suggest La Niña-like or ENSO-neutral conditions,  
 661 linked to drier periods in central Chile (Garreaud et al., 2017, 2020). Wetter phases exhibit  
 662 slightly higher precipitation than the pre-industrial mean, whereas drier phases show  
 663 anomalies near zero for both rainfall and zonal winds (Figs. 6D and 6H). There is no  
 664 indication of a northward shift of the SWW during the Middle Holocene. Nevertheless, the  
 665 TT19-3A record shows elevated fluvial inputs and stable lake levels compared to other

666 contemporaneous basins, suggesting ATTL may have served as a hydrological refuge under  
667 regional aridity (Jenny et al., 2002, 2003).

668 Early Holocene begins with an important change in the hunter gatherer settlement pattern at  
669 the ATTL but also in the surrounding areas. Late Pleistocene logistical and/or ephemeral  
670 bases are replaced by more intensive residential camps at ca. 9.0 cal ka BP (Barberena et  
671 al., 2017; Tagle and del Río, 2005). This shift coincided with the second phase of human  
672 occupation identified at TT-3 (Labarca et al., 2024) in the lower portion of the L2 facies (ca.  
673 8.5 cal ka BP). Nevertheless, considering the available radiocarbon dates, Middle Holocene  
674 (between ca. 7.2 - 6.0 ka cal BP) was possibly the most intensive period of human  
675 occupation of the ATTL (Tagle and del Río, 2005). Human burials were added into  
676 residential areas, conforming complex mounds characterized by intensive domestic refuse  
677 and mixed stratigraphic deposits. Some isolated dates suggest that these sites were  
678 probably occupied less recurrently until the beginning of the Late Holocene (ca. 4.0 cal ka  
679 BP). At least seven of these structures were documented at the lake's shore, but only a few  
680 were systematically excavated, thus, their synchronicity, anthropological, artefactual and  
681 ecofactual particularities and their possible relations are not yet fully understood. The  
682 systematic presence of human burials in these sites stress the pivotal role that ATTL played  
683 for the hunter-gatherers during climatic oscillations, and allows us to hypothesize mobility  
684 reduction and, possibly, incipient territorialism. On a regional scale, human concentration in  
685 particularly productive locations such as the ATTL could explain the scarce record of mid-  
686 Holocene archaeological sites elsewhere in central Chile. In this scenario, the lake acts as a  
687 refuge by concentrating abundant resources such as water, food, and raw materials,  
688 especially during periods of prolonged drought (Barberena et al., 2017; Núñez and Grosjean,  
689 2003).

## 690 5. Conclusions

691  
692 The ATTL basin in central Chile offers a detailed record of climatic and human dynamics  
693 during the Late Pleistocene and the beginning of the Holocene. The integration of  
694 stratigraphic, geochemical, and paleoecological data highlights the role of regional  
695 hydroclimatic changes in guiding early human mobility, resource use, and settlement  
696 choices.

697 Coupling local lacustrine evidence with large-scale climate models (PMIP4-CMIP6) shows  
698 that regional drivers (SWW, SPSA, STJ, ENSO-like phenomena) manifested in a highly  
699 localized manner. This underscores the importance of linking local data sets to global

700 circulation patterns to avoid oversimplification of paleoclimate scenarios. During the Late  
701 Pleistocene, high-energy fluvial inflows driven by atmospheric shifts (STJ intensity, ITCZ,  
702 and SWW displacement) occurred, but these changes did not significantly impact the lake's  
703 water balance, indicating a complex hydrological regime needing further study. These shifts  
704 coincide with global climate phenomena such as the HS1, B/A, and the CAPE and ACR  
705 events in South America.

706 The Pleistocene-Holocene transition brought warmer, wetter conditions, driving rapid  
707 environmental changes and the development of new habitats in the lake and its watershed.  
708 The clear association of human presence with megafaunal remains and *Sporormiella* spores  
709 indicates that shifting shoreline habitats, prey distribution, and hunter-gatherer activities  
710 converged at ATTL. The TT19-3A record provides valuable insights into how climate and  
711 humans jointly influenced the extinction and dispersal of Late Pleistocene fauna. Rapid lake-  
712 level fluctuations and episodes of high-energy deposition during the Late Pleistocene and  
713 Holocene serve as analogs for modern climate variability. Historical ecology, including  
714 research on safe environmental enclaves and the reorganization of human settlements  
715 during extreme climate events, provides valuable perspectives on resilience strategies for  
716 contemporary and future water resource planning.

717 Evidence from the ATTL basin indicates that, after a humid Early Holocene phase, increased  
718 depositional variability and shoreline habitat changes were driven by ENSO-like shifts.  
719 However, during the Middle Holocene, the ATTL shows relatively stable lake levels —  
720 contrasting with other central Chilean sites— thereby providing favorable conditions for  
721 humans and fauna. This hydrological stability, coupled with abundant resources, suggests  
722 that the ATTL basin functioned as an ecological refuge during drought periods. The  
723 intensification of occupation, formation of mounds, and emergence of more complex social  
724 practices (e.g., burials) reflect how favorable local conditions could catalyze socio-cultural  
725 development.

726 The TT19-3A sequence underscores the strong interplay between climate and human  
727 history in central Chile, showing how early hunter-gatherers adapted to periods of both  
728 abundance and stress. While the “ecorefugio” model derived from this study provides  
729 valuable insights, caution is warranted when applying it to other South American contexts,  
730 where variations in geology, topography, and climate may yield distinct sedimentological and  
731 ecological outcomes. Although robust age-depth models have been used, certain  
732 stratigraphic units —particularly those from the late Pleistocene— remain less densely

733 dated, potentially obscuring abrupt environmental fluctuations and complicating the  
734 correlation of short-lived events.

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## 745 7. Data availability

746 All code for data processing, written in Python and R, is publicly available on Zenodo and on  
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## 755 9. Author contributions

756 MFA conceived the study and wrote the manuscript with contributions from all coauthors.

757 Conceptualization: MFA, RL.

758 Methodology: MFA, RL, CGA, JA, MB, LAV, ADH, BVG, JI.

759 Investigation: MFA, RL, CGA, JA, MB, LAV, ADH, BVG, CL, EGG, NV.

760 Visualization: MFA, MB, LAV.

761 Supervision: MFA, RL.

762 Writing—original draft: MFA.

763 Writing—review & editing: MFA, RL, CGA, JA, MB, LAV, ADH, BVG, JI, CL, EGG.

764 10. Competing interests

765 The authors declare that they have no competing interests.

## 767 11. References

- 768 Alley, R.B., 2000. The Younger Dryas cold interval as viewed from central Greenland. *Quat.*  
769 *Sci. Rev.* 19, 213–226. [https://doi.org/10.1016/S0277-3791\(99\)00062-1](https://doi.org/10.1016/S0277-3791(99)00062-1)
- 770 Amann, B., Bertrand, S., Alvarez-Garretón, C., Reid, B., 2022. Seasonal Variations in Fjord  
771 Sediment Grain Size: A Pre-requisite for Hydrological and Climate Reconstructions in  
772 Partially Glacierized Watersheds (Baker River, Patagonia). *J. Geophys. Res. Earth*  
773 *Surf.* 127, e2021JF006391. <https://doi.org/10.1029/2021JF006391>
- 774 Andermann, T., Faurby, S., Turvey, S.T., Antonelli, A., Silvestro, D., 2020. The past and  
775 future human impact on mammalian diversity. *Sci. Adv.* 6, eabb2313.  
776 <https://doi.org/10.1126/sciadv.abb2313>
- 777 Barberena, R., Méndez, C., de Porras, M.E., 2017. Zooming out from archaeological  
778 discontinuities: The meaning of mid-Holocene temporal troughs in South American  
779 deserts. *J. Anthropol. Archaeol.*, *Archaeological discontinuities: Comparative*  
780 *perspectives for the southern hemisphere* 46, 68–81.  
781 <https://doi.org/10.1016/j.jaa.2016.07.003>
- 782 Barnosky, A.D., Koch, P.L., Feranec, R.S., Wing, S.L., Shabel, A.B., 2004. Assessing the  
783 Causes of Late Pleistocene Extinctions on the Continents. *Science* 306, 70–75.  
784 <https://doi.org/10.1126/science.1101476>
- 785 Baumann, C., 2023. The paleo-synanthropic niche: a first attempt to define animal's  
786 adaptation to a human-made micro-environment in the Late Pleistocene. *Archaeol.*  
787 *Anthropol. Sci.* 15, 63. <https://doi.org/10.1007/s12520-023-01764-x>
- 788 Beal, E.J., House, C.H., Orphan, V.J., 2009. Manganese- and Iron-Dependent Marine  
789 Methane Oxidation. *Science* 325, 184–187. <https://doi.org/10.1126/science.1169984>
- 790 Benito, X., Trobajo, R., Ibáñez, C., 2015. Benthic diatoms in a Mediterranean delta:  
791 ecological indicators and a conductivity transfer function for paleoenvironmental  
792 studies. *J. Paleolimnol.* 54, 171–188. <https://doi.org/10.1007/s10933-015-9845-3>
- 793 Bird, D., Miranda, L., Vander Linden, M., Robinson, E., Bocinsky, R.K., Nicholson, C.,  
794 Capriles, J.M., Finley, J.B., Gayo, E.M., Gil, A., d'Alpoim Guedes, J., Hoggarth, J.A.,  
795 Kay, A., Loftus, E., Lombardo, U., Mackie, M., Palmisano, A., Solheim, S., Kelly,  
796 R.L., Freeman, J., 2022. p3k14c, a synthetic global database of archaeological  
797 radiocarbon dates. *Sci. Data* 9, 27. <https://doi.org/10.1038/s41597-022-01118-7>
- 798 Blaauw, M., Christen, J.A., 2011. Flexible paleoclimate age-depth models using an  
799 autoregressive gamma process. *Bayesian Anal.* 6, 457–474.  
800 <https://doi.org/10.1214/11-BA618>
- 801 Borrero, L.A., 1999. The Prehistoric Exploration and Colonization of Fuego-Patagonia. *J.*  
802 *World Prehistory* 13, 321–355. <https://doi.org/10.1023/A:1022341730119>
- 803 Braje, T.J., Erlandson, J.M., 2013. Human acceleration of animal and plant extinctions: A  
804 Late Pleistocene, Holocene, and Anthropocene continuum. *Anthropocene, When*  
805 *Humans Dominated the Earth: Archeological Perspectives on the Anthropocene* 4,  
806 14–23. <https://doi.org/10.1016/j.ancene.2013.08.003>
- 807 Broecker, W.S., 1995. *The glacial world according to Wally*. Eldigio Press.
- 808 Campos, D., Rondanelli, R., 2023. ENSO-Related Precipitation Variability in Central Chile:  
809 The Role of Large Scale Moisture Transport. *J. Geophys. Res. Atmospheres* 128,  
810 e2023JD038671. <https://doi.org/10.1029/2023JD038671>
- 811 Cheng, H., Zhang, H., Spötl, C., Baker, J., Sinha, A., Li, H., Bartolomé, M., Moreno, A.,  
812 Kathayat, G., Zhao, J., Dong, X., Li, Y., Ning, Y., Jia, X., Zong, B., Ait Brahim, Y.,  
813 Pérez-Mejías, C., Cai, Y., Novello, V.F., Cruz, F.W., Severinghaus, J.P., An, Z.,  
814 Edwards, R.L., 2020. Timing and structure of the Younger Dryas event and its  
815 underlying climate dynamics. *Proc. Natl. Acad. Sci.* 117, 23408–23417.  
816 <https://doi.org/10.1073/pnas.2007869117>
- 817 Ciolkosz, E.J., Carter, B.J., Hoover, M.T., Cronsce, R.C., Waltman, W.J., Dobos, R.R., 1990.



- Genesis of soils and landscapes in the Ridge and Valley province of central Pennsylvania. *Geomorphology, Proceedings of the 21st Annual Binghamton Symposium in Geomorphology* 3, 245–261. [https://doi.org/10.1016/0169-555X\(90\)90006-C](https://doi.org/10.1016/0169-555X(90)90006-C)
- Collazo, S., García-Herrera, R., Barriopedro, D., 2024. Summer upper-level jets modulate the response of South American climate to ENSO. *Clim. Dyn.* 62, 1031–1054. <https://doi.org/10.1007/s00382-023-06955-9>
- Crema, E.R., Bevan, A., 2021. INFERENCE FROM LARGE SETS OF RADIOCARBON DATES: SOFTWARE AND METHODS. *Radiocarbon* 63, 23–39. <https://doi.org/10.1017/RDC.2020.95>
- Dansgaard, W., 1964. Stable isotopes in precipitation. *Tellus* 16, 436–468. <https://doi.org/10.3402/tellusa.v16i4.8993>
- Dietze, E., Maussion, F., Ahlborn, M., Diekmann, B., Hartmann, K., Henkel, K., Kasper, T., Lockot, G., Opitz, S., Haberzettl, T., 2014. Sediment transport processes across the Tibetan Plateau inferred from robust grain-size end members in lake sediments. *Clim. Past* 10, 91–106. <https://doi.org/10.5194/cp-10-91-2014>
- Fathi-Moghadam, M., Arman, A., Emamgholizadeh, S., Alikhani, A., 2011. Settling Properties of Cohesive Sediments in Lakes and Reservoirs. *J. Waterw. Port Coast. Ocean Eng.* 137, 204–209. [https://doi.org/10.1061/\(ASCE\)WW.1943-5460.0000078](https://doi.org/10.1061/(ASCE)WW.1943-5460.0000078)
- Fernández-Navarro, H., García, J.-L., Nussbaumer, S.U., Tikhomirov, D., Pérez, F., Gärtner-Roer, I., Christl, M., Egli, M., 2024. <sup>10</sup>Be chronology of the Last Glacial Maximum and Termination in the Andes of central Chile: The record of the Universidad Glacier (34° S). *Quat. Sci. Rev.* 344, 108968. <https://doi.org/10.1016/j.quascirev.2024.108968>
- Flores-Aqueveque, V., Rojas, M., Aguirre, C., Arias, P.A., González, C., 2020. South Pacific Subtropical High from the late Holocene to the end of the 21st century: insights from climate proxies and general circulation models. *Clim. Past* 16, 79–99. <https://doi.org/10.5194/cp-16-79-2020>
- Fritz, S.C., 2013. DIATOM METHODS | Salinity and Climate Reconstructions from Continental Lakes, in: Elias, S.A., Mock, C.J. (Eds.), *Encyclopedia of Quaternary Science (Second Edition)*. Elsevier, Amsterdam, pp. 507–515. <https://doi.org/10.1016/B978-0-444-53643-3.00221-1>
- Frugone-Alvarez, M., Latorre, C., Barreiro-Lostres, F., Giralt, S., Moreno, A., Polanco-Martínez, J., Maldonado, A., Carrevedo, M.L., Bernárdez, P., Prego, R., Delgado Huertas, A., Fuentealba, M., Valero-Garcés, B., 2020. Volcanism and climate change as drivers in Holocene depositional dynamic of Laguna del Maule (Andes of central Chile – 36° S). *Clim Past* 16, 1097–1125. <https://doi.org/10.5194/cp-16-1097-2020>
- Garreaud, R., 2007. Precipitation and Circulation Covariability in the Extratropics. *J. Clim.* 20, 4789–4797. <https://doi.org/10.1175/JCLI4257.1>
- Garreaud, R.D., Alvarez-Garretón, C., Barichivich, J., Boisier, J.P., Christie, D., Galleguillos, M., LeQuesne, C., McPhee, J., Zambrano-Bigiarini, M., 2017. The 2010–2015 megadrought in central Chile: impacts on regional hydroclimate and vegetation. *Hydrol. Earth Syst. Sci.* 21, 6307–6327. <https://doi.org/10.5194/hess-21-6307-2017>
- Garreaud, R.D., Boisier, J.P., Rondanelli, R., Montecinos, A., Sepúlveda, H.H., Veloso-Aguila, D., 2020. The Central Chile Mega Drought (2010–2018): A climate dynamics perspective. *Int. J. Climatol.* 40, 421–439. <https://doi.org/10.1002/joc.6219>
- Garreaud, R.D., Vuille, M., Compagnucci, R., Marengo, J., 2009. Present-day South American climate. *Palaeogeogr. Palaeoclimatol. Palaeoecol.*, Long-term multi-proxy climate reconstructions and dynamics in South America (LOTRED-SA): State of the art and perspectives 281, 180–195. <https://doi.org/10.1016/j.palaeo.2007.10.032>
- Gayo, E.M., Latorre, C., Jordan, T.E., Nester, P.L., Estay, S.A., Ojeda, K.F., Santoro, C.M., 2012. Late Quaternary hydrological and ecological changes in the hyperarid core of the northern Atacama Desert (~ 21°S). *Earth-Sci. Rev.* 113, 120–140. <https://doi.org/10.1016/j.earscirev.2012.04.003>
- Gillett, Z.E., Hendon, H.H., Arblaster, J.M., Lim, E.-P., 2021. Tropical and Extratropical

- 873 Influences on the Variability of the Southern Hemisphere Wintertime Subtropical Jet.  
874 J. Clim. 34, 4009–4022. <https://doi.org/10.1175/JCLI-D-20-0460.1>
- 875 Gonfiantini, R., Roche, M.-A., Olivry, J.-C., Fontes, J.-C., Zuppi, G.M., 2001. The altitude  
876 effect on the isotopic composition of tropical rains. Chem. Geol. 181, 147–167.  
877 [https://doi.org/10.1016/S0009-2541\(01\)00279-0](https://doi.org/10.1016/S0009-2541(01)00279-0)
- 878 González-Pinilla, F.J., Latorre, C., Rojas, M., Houston, J., Rocuant, M.I., Maldonado, A.,  
879 Santoro, C.M., Quade, J., Betancourt, J.L., 2021. High- and low-latitude forcings  
880 drive Atacama Desert rainfall variations over the past 16,000 years. Sci. Adv. 7,  
881 eabg1333. <https://doi.org/10.1126/sciadv.abg1333>
- 882 Gutiérrez, E.G., Stockli, D.F., 2023. Fluvial response to Late Pleistocene–Holocene climate  
883 change in the Colorado River drainage, central Texas, USA. Geology 51, 929–934.  
884 <https://doi.org/10.1130/G51239.1>
- 885 Hall, B., 2016. Facies classification using machine learning. Lead. Edge 35, 906–909.  
886 <https://doi.org/10.1190/tle35100906.1>
- 887 Herrera-Ossandón, M., Easton, G., Antinao, J.L., Forman, S.L., 2023. Late Quaternary  
888 glacier advances in the Andes of Santiago, central Chile, and paleoclimatic  
889 implications. Front. Earth Sci. 11. <https://doi.org/10.3389/feart.2023.1192812>
- 890 Heusser, C.J., 1983. Quaternary Pollen Record from Laguna de Tagua Tagua, Chile.  
891 Science 219, 1429–1432. <https://doi.org/10.1126/science.219.4591.1429>
- 892 Heusser, C.J., 1971. Pollen and spores of Chile. Modern types of the Pteridophyta,  
893 Gymnospermae, and Angiospermae. University of Arizona Press.
- 894 Hogg, A.G., Heaton, T.J., Hua, Q., Palmer, J.G., Turney, C.S., Southon, J., Bayliss, A.,  
895 Blackwell, P.G., Boswijk, G., Ramsey, C.B., Pearson, C., Petchey, F., Reimer, P.,  
896 Reimer, R., Wacker, L., 2020. SHCal20 Southern Hemisphere Calibration, 0–55,000  
897 Years cal BP. Radiocarbon 62, 759–778. <https://doi.org/10.1017/RDC.2020.59>
- 898 Horton, T.W., Defliese, W.F., Tripathi, A.K., Oze, C., 2016. Evaporation induced 18O and 13C  
899 enrichment in lake systems: A global perspective on hydrologic balance effects.  
900 Quat. Sci. Rev., Water Isotope Systematics 131, 365–379.  
901 <https://doi.org/10.1016/j.quascirev.2015.06.030>
- 902 International Committee for Phytolith Taxonomy (ICPT), 2019. International Code for  
903 Phytolith Nomenclature (ICPN) 2.0. Ann. Bot. 124, 189–199.  
904 <https://doi.org/10.1093/aob/mcz064>
- 905 Jara-Arancio, P., da Silva Carvalho, C., Carmona-Ortiz, M.R., Bustamante, R.O., Schmidt  
906 Villela, P.M., da Silva Andrade, S.C., Peña-Gómez, F.T., González, L.A., Fleury, M.,  
907 2022. Genetic Diversity and Population Structure of *Jubaea chilensis*, an Endemic  
908 and Monotype Gender from Chile, Based on SNP Markers. Plants 11, 1959.  
909 <https://doi.org/10.3390/plants11151959>
- 910 Jenny, B., Valero-Garcés, B.L., Villa-Martinez, R., Urrutia, R., Geyh, M., Veit, H., 2002. Early  
911 to Mid-Holocene Aridity in Central Chile and the Southern Westerlies: The Laguna  
912 Aculeo Record (34°S). Quat. Res. 58, 160–170.  
913 <https://doi.org/10.1006/qres.2002.2370>
- 914 Jenny, B., Wilhelm, D., Valero-Garcés, B., 2003. The Southern Westerlies in Central Chile:  
915 Holocene precipitation estimates based on a water balance model for Laguna Aculeo  
916 (33 50' S). Clim. Dyn. 20, 269–280.
- 917 Kaiser, J., Lamy, F., Hebbeln, D., 2005. A 70-kyr sea surface temperature record off  
918 southern Chile (Ocean Drilling Program Site 1233). Paleoceanography 20.  
919 <https://doi.org/10.1029/2005PA001146>
- 920 Kaiser, J., Schefuß, E., Collins, J., Garreaud, R., Stuut, J.-B.W., Ruggieri, N., De Pol-Holz,  
921 R., Lamy, F., 2024. Orbital modulation of subtropical versus subantarctic moisture  
922 sources in the southeast Pacific mid-latitudes. Nat. Commun. 15, 7512.  
923 <https://doi.org/10.1038/s41467-024-51985-4>
- 924 Kaiser, J., Schefuß, E., Lamy, F., Mohtadi, M., Hebbeln, D., 2008. Glacial to Holocene  
925 changes in sea surface temperature and coastal vegetation in north central Chile:  
926 high versus low latitude forcing. Quat. Sci. Rev. 27, 2064–2075.
- 927 Khider, D., Emile-Geay, J., Zhu, F., James, A., Landers, J., Ratnakar, V., Gil, Y., 2022.

- Pyleoclim: Paleoclimate Timeseries Analysis and Visualization With Python. *Paleoceanogr. Paleoclimatology* 37, e2022PA004509. <https://doi.org/10.1029/2022PA004509>
- Kilian, R., Lamy, F., 2012. A review of Glacial and Holocene paleoclimate records from southernmost Patagonia (49–55°S). *Quat. Sci. Rev.* 53, 1–23. <https://doi.org/10.1016/j.quascirev.2012.07.017>
- Kim, J.-H., Schneider, R.R., Hebbeln, D., Müller, P.J., Wefer, G., 2002. Last deglacial sea-surface temperature evolution in the Southeast Pacific compared to climate changes on the South American continent. *Quat. Sci. Rev.* 21, 2085–2097. [https://doi.org/10.1016/S0277-3791\(02\)00012-4](https://doi.org/10.1016/S0277-3791(02)00012-4)
- Labarca, R., Frugone-Álvarez, M., Vilches, L., Blanco, J.F., Peñaloza, Á., Godoy-Aguirre, C., Lizama-Catalán, Á., Oyarzo, C., Tornero, C., González-Guarda, E., Delgado, A., Sepúlveda, M., Soto-Huenchuman, P., 2024. Taguatagua 3: A new late Pleistocene settlement in a highly suitable lacustrine habitat in central Chile (34°S). *PLOS ONE* 19, e0302465. <https://doi.org/10.1371/journal.pone.0302465>
- Lamy, F., Hebbeln, D., Wefer, G., 1999. High-Resolution Marine Record of Climatic Change in Mid-latitude Chile during the Last 28,000 Years Based on Terrigenous Sediment Parameters. *Quat. Res.* 51, 83–93. <https://doi.org/10.1006/qres.1998.2010>
- Lamy, F., Kilian, R., Arz, H.W., Francois, J.-P., Kaiser, J., Prange, M., Steinke, T., 2010. Holocene changes in the position and intensity of the southern westerly wind belt. *Nat. Geosci.* 3, 695–699. <https://doi.org/10.1038/ngeo959>
- Latorre, C., Betancourt, J.L., Arroyo, M.T.K., 2006. Late Quaternary vegetation and climate history of a perennial river canyon in the Río Salado basin (22°S) of Northern Chile. *Quat. Res.* 65, 450–466. <https://doi.org/10.1016/j.yqres.2006.02.002>
- León-Lobos, P., Ellis, R.H., 2005. Seed survival in Chilean *Nothofagus* in response to desiccation and storage. *Seed Sci. Res.* 15, 113–123. <https://doi.org/10.1079/SSR2005204>
- Li, S., Liu, W., 2022. Deciphering the Migration of the Intertropical Convergence Zone During the Last Deglaciation. *Geophys. Res. Lett.* 49, e2022GL098806. <https://doi.org/10.1029/2022GL098806>
- Liu, Z., Otto-Bliesner, B.L., He, F., Brady, E.C., Tomas, R., Clark, P.U., Carlson, A.E., Lynch-Stieglitz, J., Curry, W., Brook, E., Erickson, D., Jacob, R., Kutzbach, J., Cheng, J., 2009. Transient Simulation of Last Deglaciation with a New Mechanism for Bølling-Allerød Warming. *Science* 325, 310–314. <https://doi.org/10.1126/science.1171041>
- Lizama-Catalán, Á., Labarca, R., 2023. Who eats What: Unravelling a complex taphonomic scenario in the lacustrine deposits of the late Pleistocene archaeological site, Taguatagua 1, central Chile. *Quat. Sci. Rev.* 300, 107831. <https://doi.org/10.1016/j.quascirev.2022.107831>
- Lu, J., Chen, G., Frierson, D.M.W., 2008. Response of the Zonal Mean Atmospheric Circulation to El Niño versus Global Warming. *J. Clim.* 21, 5835–5851. <https://doi.org/10.1175/2008JCLI2200.1>
- Magri, D., Di Rita, F., 2015. Archaeopalynological Preparation Techniques, in: Yeung, E.C.T., Stasolla, C., Sumner, M.J., Huang, B.Q. (Eds.), *Plant Microtechniques and Protocols*. Springer International Publishing, Cham, pp. 495–506. [https://doi.org/10.1007/978-3-319-19944-3\\_27](https://doi.org/10.1007/978-3-319-19944-3_27)
- Maldonado, A., Méndez, C., Ugalde, P., Jackson, D., Seguel, R., Latorre, C., 2010. Early Holocene climate change and human occupation along the semiarid coast of north-central Chile. *J. Quat. Sci.* 25, 985–988. <https://doi.org/10.1002/jqs.1385>
- Malhi, Y., Doughty, C.E., Galetti, M., Smith, F.A., Svenning, J.-C., Terborgh, J.W., 2016. Megafauna and ecosystem function from the Pleistocene to the Anthropocene. *Proc. Natl. Acad. Sci.* 113, 838–846. <https://doi.org/10.1073/pnas.1502540113>
- Mann, D.H., Groves, P., Gaglioti, B.V., Shapiro, B.A., 2019. Climate-driven ecological stability as a globally shared cause of Late Quaternary megafaunal extinctions: the Plaids and Stripes Hypothesis. *Biol. Rev.* 94, 328–352. <https://doi.org/10.1111/brv.12456>

- 983 Martin, L.C.P., Blard, P.-H., Lavé, J., Condom, T., Prémaillon, M., Jomelli, V., Brunstein, D.,  
 984 Lupker, M., Charreau, J., Mariotti, V., Tibari, B., ASTER Team, Davy, E., 2018. Lake  
 985 Tauca highstand (Heinrich Stadial 1a) driven by a southward shift of the Bolivian  
 986 High. *Sci. Adv.* 4, eaar2514. <https://doi.org/10.1126/sciadv.aar2514>
- 987 Mathiasen, P., Venegas-González, A., Fresia, P., Premoli, A.C., 2020. A relic of the past:  
 988 current genetic patterns of the palaeoendemic tree *Nothofagus macrocarpa* were  
 989 shaped by climatic oscillations in central Chile. *Ann. Bot.* 126, 891–904.  
 990 <https://doi.org/10.1093/aob/mcaa111>
- 991 Mayle, F.E., Iriarte, J., 2014. Integrated palaeoecology and archaeology – a powerful  
 992 approach for understanding pre-Columbian Amazonia. *J. Archaeol. Sci.*, The world  
 993 reshaped: practices and impacts of early agrarian societies 51, 54–64.  
 994 <https://doi.org/10.1016/j.jas.2012.08.038>
- 995 McCormack, J., Nehrke, G., Jöns, N., Immenhauser, A., Kwiecien, O., 2019. Refining the  
 996 interpretation of lacustrine carbonate isotope records: Implications of a mineralogy-  
 997 specific Lake Van case study. *Chem. Geol.* 513, 167–183.  
 998 <https://doi.org/10.1016/j.chemgeo.2019.03.014>
- 999 McCrea, J.M., 1950. On the Isotopic Chemistry of Carbonates and a Paleotemperature  
 1000 Scale. *J. Chem. Phys.* 18, 849–857. <https://doi.org/10.1063/1.1747785>
- 1001 McKay, N.P., Emile-Geay, J., Khider, D., 2021. geoChronR – an R package to model,  
 1002 analyze, and visualize age-uncertain data. *Geochronology* 3, 149–169.  
 1003 <https://doi.org/10.5194/gchron-3-149-2021>
- 1004 Méndez, C., Jackson, D., 2015. Terminal pleistocene lithic technology and use of space in  
 1005 Central Chile. *Chungara Rev. Antropol. Chil.* 47, 53–65.
- 1006 Montané, J., 1969. Fechado del nivel superior de Tagua Tagua. *Not. Mens. Mus. Nac. Hist.*  
 1007 *Nat.* 14, 9–10.
- 1008 Montané, J., 1968. Paleo-Indian Remains from Laguna de Tagua Tagua, Central Chile.  
 1009 *Science* 161, 1137–1138. <https://doi.org/10.1126/science.161.3846.1137>
- 1010 Morales, E.A., Siver, P.A., Trainor, F.R., 2001. Identification of diatoms (Bacillariophyceae)  
 1011 during ecological assessments: Comparison between Light Microscopy and  
 1012 Scanning Electron Microscopy techniques. *Proc. Acad. Nat. Sci. Phila.* 151, 95–103.  
 1013 [https://doi.org/10.1635/0097-3157\(2001\)151\[0095:IODBDE\]2.0.CO;2](https://doi.org/10.1635/0097-3157(2001)151[0095:IODBDE]2.0.CO;2)
- 1014 Moreno, P.I., Francois, J.P., Moy, C.M., Villa-Martínez, R., 2010. Covariability of the  
 1015 Southern Westerlies and atmospheric CO<sub>2</sub> during the Holocene. *Geology* 38, 727–  
 1016 730. <https://doi.org/10.1130/G30962.1>
- 1017 Moreno, P.I., León, A.L., 2003. Abrupt vegetation changes during the last glacial to  
 1018 Holocene transition in mid-latitude South America. *J. Quat. Sci.* 18, 787–800.  
 1019 <https://doi.org/10.1002/jqs.801>
- 1020 Moreno-Mayar, J.V., Vinner, L., Damgaard, P. de B., Fuente, C. de la, Chan, J., Spence,  
 1021 J.P., Allentoft, M.E., Vimala, T., Racimo, F., Pinotti, T., Rasmussen, S., Margaryan,  
 1022 A., Orbegozo, M.I., Myopotamitaki, D., Wooller, M., Bataille, C., Becerra-Valdivia, L.,  
 1023 Chivall, D., Comeskey, D., Deviese, T., Grayson, D.K., George, L., Harry, H.,  
 1024 Alexandersen, V., Primeau, C., Erlandson, J., Rodrigues-Carvalho, C., Reis, S.,  
 1025 Bastos, M.Q.R., Cybulski, J., Vullo, C., Morello, F., Vilar, M., Wells, S., Gregersen,  
 1026 K., Hansen, K.L., Lynnerup, N., Lahr, M.M., Kjær, K., Strauss, A., Alfonso-Durruty,  
 1027 M., Salas, A., Schroeder, H., Higham, T., Malhi, R.S., Rasic, J.T., Souza, L., Santos,  
 1028 F.R., Malaspinas, A.-S., Sikora, M., Nielsen, R., Song, Y.S., Meltzer, D.J., Willerslev,  
 1029 E., 2018. Early human dispersals within the Americas. *Science* 362.  
 1030 <https://doi.org/10.1126/science.aav2621>
- 1031 Mottl, O., Flantua, S.G.A., Bhatta, K.P., Felde, V.A., Giesecke, T., Goring, S., Grimm, E.C.,  
 1032 Haberle, S., Hooghiemstra, H., Ivory, S., Kuneš, P., Wolters, S., Seddon, A.W.R.,  
 1033 Williams, J.W., 2021. Global acceleration in rates of vegetation change over the past  
 1034 18,000 years. *Science* 372, 860–864. <https://doi.org/10.1126/science.abg1685>
- 1035 Moy, C.M., Seltzer, G.O., Rodbell, D.T., Anderson, D.M., 2002. Variability of El  
 1036 Niño/Southern Oscillation activity at millennial timescales during the Holocene epoch.  
 1037 *Nature* 420, 162–165. <https://doi.org/10.1038/nature01194>

- Novello, V.F., Cruz, F.W., Vuille, M., Strikis, N.M., Edwards, R.L., Cheng, H., Emerick, S., de Paula, M.S., Li, X., Barreto, E. de S., Karmann, I., Santos, R.V., 2017. A high-resolution history of the South American Monsoon from Last Glacial Maximum to the Holocene. *Sci. Rep.* 7, 44267. <https://doi.org/10.1038/srep44267>
- Núñez, L., Varela, J., Casamiquela, R., Schiappacasse, V., Niemeyer, H., & Villagrán, C., 1994. Taguatagua Basin in Chile: The Late Pleistocene environment and human occupations. *Revista Chilena de Historia Natural*, 67(3), 503–519.
- Núñez, L., Grosjean, M., 2003. Biodiversity and Human Impact During the Last 11,000 Years in North-Central Chile, in: Bradshaw, G.A., Marquet, P.A. (Eds.), *How Landscapes Change: Human Disturbance and Ecosystem Fragmentation in the Americas*. Springer, Berlin, Heidelberg, pp. 7–17. [https://doi.org/10.1007/978-3-662-05238-9\\_2](https://doi.org/10.1007/978-3-662-05238-9_2)
- Núñez, L., Varela, J., Casamiquela, R., Villagrán, C., 1994. Reconstrucción Multidisciplinaria de la Ocupación Prehistórica de Quereo, Centro de Chile. *Lat. Am. Antiq.* 5, 99–118. <https://doi.org/10.2307/971558>
- Orellana, H., Latorre, C., García, J.-L., Lambert, F., 2023. Spatial analysis of paleoclimate variations based on proxy records in the south-central Andes (18°–35° S) from 32 to 4 ka. *Quat. Sci. Rev.* 313, 108174. <https://doi.org/10.1016/j.quascirev.2023.108174>
- Ortega, C., Vargas, G., Rutllant, J.A., Jackson, D., Méndez, C., 2012. Major hydrological regime change along the semiarid western coast of South America during the early Holocene. *Quat. Res.* 78, 513–527. <https://doi.org/10.1016/j.yqres.2012.08.002>
- Orton, G.J., Reading, H.G., 1993. Variability of deltaic processes in terms of sediment supply, with particular emphasis on grain size. *Sedimentology* 40, 475–512. <https://doi.org/10.1111/j.1365-3091.1993.tb01347.x>
- Parnell, A.C., Haslett, J., Allen, J.R.M., Buck, C.E., Huntley, B., 2008. A flexible approach to assessing synchronicity of past events using Bayesian reconstructions of sedimentation history. *Quat. Sci. Rev.* 27, 1872–1885. <https://doi.org/10.1016/j.quascirev.2008.07.009>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É., 2011. Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.
- Pedro, J.B., Bostock, H.C., Bitz, C.M., He, F., Vandergoes, M.J., Steig, E.J., Chase, B.M., Krause, C.E., Rasmussen, S.O., Markle, B.R., Cortese, G., 2016. The spatial extent and dynamics of the Antarctic Cold Reversal. *Nat. Geosci.* 9, 51–55. <https://doi.org/10.1038/ngeo2580>
- Piperno, D.R., 2006. *Phytoliths. In A Comprehensive Guide for Archaeologists and Palaeoecologists*. Altamira Press, USA, New York.
- Piperno, D.R., 1988. *Phytolith Analysis. An Archaeological and Geological Perspective*. 5 - laboratory techniques, in: Piperno, D.R. (Ed.), *Phytolith Analysis*. Academic Press, San Diego, pp. 119–130. <https://doi.org/10.1016/B978-0-12-557175-3.50009-X>
- Prates, L., Perez, S.I., 2021. Late Pleistocene South American megafaunal extinctions associated with rise of Fishtail points and human population. *Nat. Commun.* 12, 2175. <https://doi.org/10.1038/s41467-021-22506-4>
- Prates, L., Politis, G.G., Perez, S.I., 2020. Rapid radiation of humans in South America after the last glacial maximum: A radiocarbon-based study. *PLOS ONE* 15, e0236023. <https://doi.org/10.1371/journal.pone.0236023>
- Quade, J., Rech, J.A., Betancourt, J.L., Latorre, C., Quade, B., Rylander, K.A., Fisher, T., 2008. Paleowetlands and regional climate change in the central Atacama Desert, northern Chile. *Quat. Res.* 69, 343–360. <https://doi.org/10.1016/j.yqres.2008.01.003>
- Rein, B., Lückge, A., Reinhardt, L., Sirocko, F., Wolf, A., Dullo, W.-C., 2005. El Niño variability off Peru during the last 20,000 years. *Paleoceanography* 20. <https://doi.org/10.1029/2004PA001099>
- Reyes-Díaz, M., Alberdi, M., Piper, F., Bravo, L.A., Corcuera, L.J., 2005. Low temperature responses of *Nothofagus dombeyi* and *Nothofagus nitida*, two evergreen species from south central Chile. *Tree Physiol.* 25, 1389–1398.

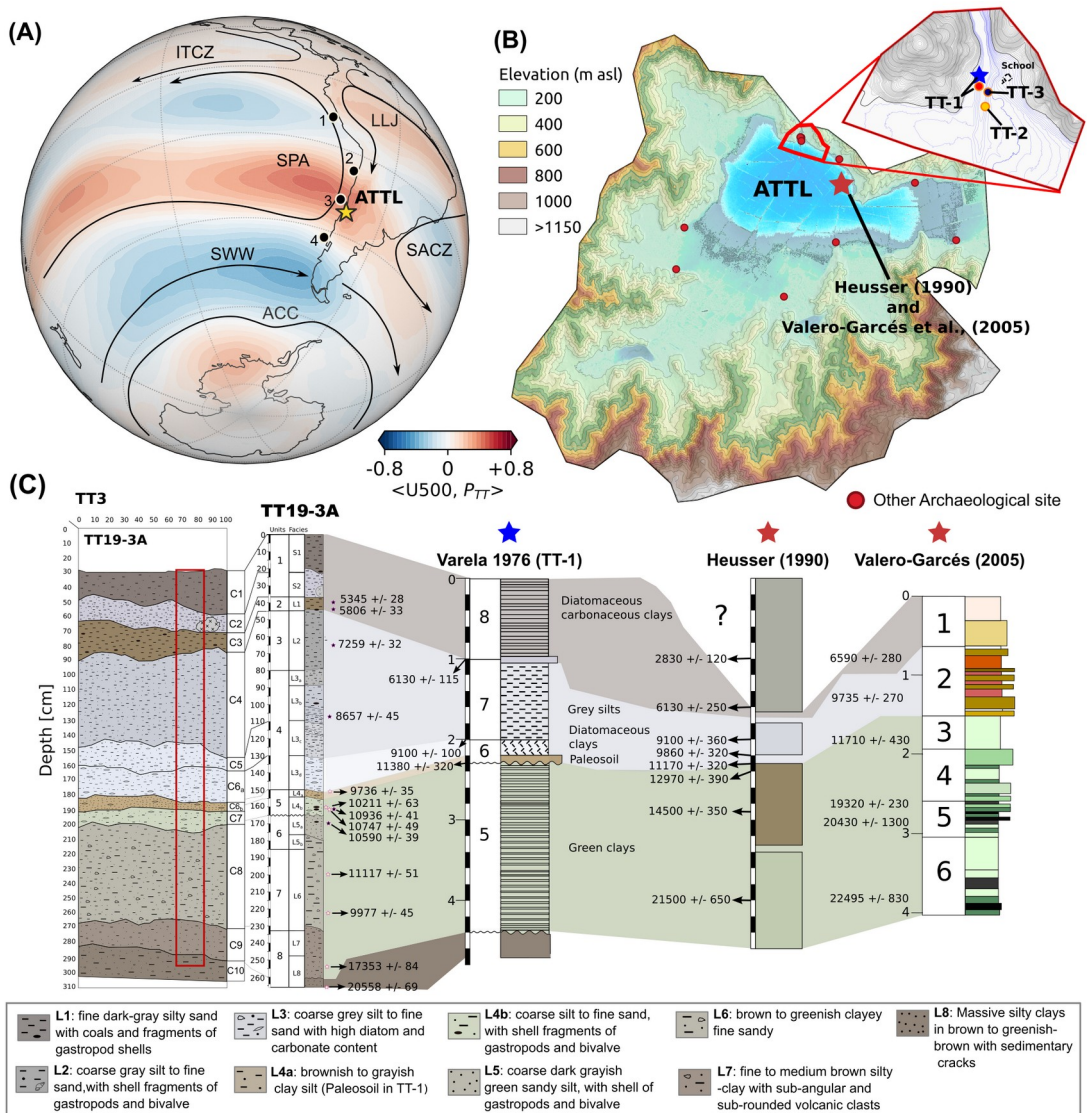
- 1093 <https://doi.org/10.1093/treephys/25.11.1389>
- 1094 Robinson, M., Morcote-Rios, G., Aceituno, F.J., Roberts, P., Berrío, J.C., Iriarte, J., 2021.
- 1095 'Moving South': Late Pleistocene Plant Exploitation and the Importance of Palm in
- 1096 the Colombian Amazon. *Quaternary* 4, 26. <https://doi.org/10.3390/quat4030026>
- 1097 Rozanski, K., Araguás-Araguás, L., Gonfiantini, R., 1993. Isotopic patterns in modern global
- 1098 precipitation. *Clim. Change Cont. Isot. Rec.* 78, 1–36.
- 1099 Sandweiss, D.H., Andrus, C.F.T., Kelley, A.R., Maasch, K.A., Reitz, E.J., Roscoe, P.B.,
- 1100 2020. Archaeological climate proxies and the complexities of reconstructing
- 1101 Holocene El Niño in coastal Peru. *Proc. Natl. Acad. Sci.* 117, 8271–8279.
- 1102 <https://doi.org/10.1073/pnas.1912242117>
- 1103 Schneider, T., Bischoff, T., Haug, G.H., 2014. Migrations and dynamics of the intertropical
- 1104 convergence zone. *Nature* 513, 45–53. <https://doi.org/10.1038/nature13636>
- 1105 Scott, W.P., Contreras, S., Bowen, G.J., Arnold, T.E., Bustamante-Ortega, R., Werne, J.P.,
- 1106 2021. Lake water based isoscape in central-south Chile reflects meteoric water. *Sci.*
- 1107 *Rep.* 11, 8725. <https://doi.org/10.1038/s41598-021-87566-4>
- 1108 Singer, B., Hildreth, W., Vincze, Y., 2000. <sup>40</sup>Ar/<sup>39</sup>Ar evidence for early deglaciation of the
- 1109 central Chilean Andes. *Geophys. Res. Lett.* 27, 1663–1666.
- 1110 <https://doi.org/10.1029/1999GL011065>
- 1111 Soteres, R.L., Sagredo, E.A., Kaplan, M.R., Martini, M.A., Moreno, P.I., Reynhout, S.A.,
- 1112 Schwartz, R., Schaefer, J.M., 2022. Glacier fluctuations in the northern Patagonian
- 1113 Andes (44°S) imply wind-modulated interhemispheric in-phase climate shifts during
- 1114 Termination 1. *Sci. Rep.* 12, 10842. <https://doi.org/10.1038/s41598-022-14921-4>
- 1115 Starratt, S.W., Kirby, M.E., Glover, K., 2021. Diatom Record of Holocene Moisture Variability
- 1116 in the San Bernardino Mountains, California, USA, in: Rosen, M.R., Finkelstein, D.B.,
- 1117 Park Boush, L., Pla-Pueyo, S. (Eds.), *Limnogeology: Progress, Challenges and*
- 1118 *Opportunities: A Tribute to Elizabeth Gierlowski-Kordesch*. Springer International
- 1119 Publishing, Cham, pp. 329–365. [https://doi.org/10.1007/978-3-030-66576-0\\_11](https://doi.org/10.1007/978-3-030-66576-0_11)
- 1120 Stenger-Kovács, C., Béres, V.B., Buczkó, K., Tapolczai, K., Padisák, J., Selmečzy, G.B.,
- 1121 Lengyel, E., 2023. Diatom community response to inland water salinization: a review.
- 1122 *Hydrobiologia* 850, 4627–4663. <https://doi.org/10.1007/s10750-023-05167-w>
- 1123 Sun, D., Bloemendal, J., Rea, D.K., Vandenberghe, J., Jiang, F., An, Z., Su, R., 2002. Grain-
- 1124 size distribution function of polymodal sediments in hydraulic and aeolian
- 1125 environments, and numerical partitioning of the sedimentary components. *Sediment.*
- 1126 *Geol.* 152, 263–277. [https://doi.org/10.1016/S0037-0738\(02\)00082-9](https://doi.org/10.1016/S0037-0738(02)00082-9)
- 1127 Tagle, B., del Río, C., 2005. Informe de prospecciones arqueológicas y excavación de
- 1128 túmulos funerarios en las provincias de Colchagua y Cachapoal. *Inf. Al Cons.*
- 1129 *Monum. Nac. Ms.*
- 1130 Terzer-Wassmuth, S., Wassenaar, L.I., Welker, J.M., Araguás-Araguás, L.J., 2021.
- 1131 Improved high-resolution global and regionalized isoscapes of  $\delta^{18}\text{O}$ ,  $\delta^2\text{H}$  and d-
- 1132 excess in precipitation. *Hydrol. Process.* 35, e14254.
- 1133 <https://doi.org/10.1002/hyp.14254>
- 1134 Trepel, J., le Roux, E., Abraham, A.J., Buitenwerf, R., Kamp, J., Kristensen, J.A., Tietje, M.,
- 1135 Lundgren, E.J., Svenning, J.-C., 2024. Meta-analysis shows that wild large
- 1136 herbivores shape ecosystem properties and promote spatial heterogeneity. *Nat. Ecol.*
- 1137 *Evol.* 8, 705–716. <https://doi.org/10.1038/s41559-024-02327-6>
- 1138 Ugalde, P.C., Joly, D., Latorre, C., Gayo, E.M., Labarca, R., Simunovic, M., McRostie, V.,
- 1139 Holliday, V.T., Quade, J., Santoro, C.M., 2024. The first peoples of the Atacama
- 1140 Desert lived among the trees: A 11,600- to 11,200-year-old grove and congregation
- 1141 site. *Proc. Natl. Acad. Sci.* 121, e2320506121.
- 1142 <https://doi.org/10.1073/pnas.2320506121>
- 1143 Valero-Garcés, B.L., Jenny, B., Rondanelli, M., Delgado-Huertas, A., Burns, S.J., Veit, H.,
- 1144 Moreno, A., 2005. Palaeohydrology of Laguna de Tagua Tagua (34° 30' S) and
- 1145 moisture fluctuations in Central Chile for the last 46 000 yr. *J. Quat. Sci.* 20, 625–
- 1146 641. <https://doi.org/10.1002/jqs.988>
- 1147 van der Vegt, H., Storms, J.E.A., Walstra, D.-J.R., Nordahl, K., Howes, N.C., Martinius,

- A.W., 2020. Grain size fractionation by process-driven sorting in sandy to muddy deltas. *Depositional Rec.* 6, 217–235. <https://doi.org/10.1002/dep2.85>
- van IJzendoorn, C.O., Hallin, C., Reniers, A.J.H.M., de Vries, S., 2023. Modeling Multi-Fraction Coastal Aeolian Sediment Transport With Horizontal and Vertical Grain-Size Variability. *J. Geophys. Res. Earth Surf.* 128, e2023JF007155. <https://doi.org/10.1029/2023JF007155>
- Vandenberghe, J., 2013. Grain size of fine-grained windblown sediment: A powerful proxy for process identification. *Earth-Sci. Rev.* 121, 18–30. <https://doi.org/10.1016/j.earscirev.2013.03.001>
- Vandenberghe, J., Sun, Y., Wang, X., Abels, H.A., Liu, X., 2018. Grain-size characterization of reworked fine-grained aeolian deposits. *Earth-Sci. Rev.* 177, 43–52. <https://doi.org/10.1016/j.earscirev.2017.11.005>
- Varela, J., 1976. Geología del Cuaternario de Laguna de Taguatagua (Provincia de O'Higgins), in: *Congreso Geológico Chileno*. pp. D81–D114.
- Villacís, L.A., Moreno, P.I., Vilanova, I., Henríquez, C.A., Henríquez, W.I., Villa-Martínez, R.P., Sepúlveda-Zúñiga, E.A., Maidana, N.I., 2023. A freshwater diatom perspective on the evolution of the southern westerlies for the past ~14,000 years in southwestern Patagonia. *Quat. Sci. Rev.* 301, 107929. <https://doi.org/10.1016/j.quascirev.2022.107929>
- Villavicencio, N.A., Lindsey, E.L., Martin, F.M., Borrero, L.A., Moreno, P.I., Marshall, C.R., Barnosky, A.D., 2016. Combination of humans, climate, and vegetation change triggered Late Quaternary megafauna extinction in the Última Esperanza region, southern Patagonia, Chile. *Ecography* 39, 125–140. <https://doi.org/10.1111/ecog.01606>
- Wang, X., Xie, G.-J., Tian, N., Dang, C.-C., Cai, C., Ding, J., Liu, B.-F., Xing, D.-F., Ren, N.-Q., Wang, Q., 2022. Anaerobic microbial manganese oxidation and reduction: A critical review. *Sci. Total Environ.* 822, 153513. <https://doi.org/10.1016/j.scitotenv.2022.153513>
- Wang, Y., Shipley, B.R., Lauer, D.A., Pineau, R.M., McGuire, J.L., 2020. Plant biomes demonstrate that landscape resilience today is the lowest it has been since end-Pleistocene megafaunal extinctions. *Glob. Change Biol.* 26, 5914–5927. <https://doi.org/10.1111/gcb.15299>
- Zhang, X., Zhou, A., Wang, X., Song, M., Zhao, Y., Xie, H., Russell, J.M., Chen, F., 2018. Unmixing grain-size distributions in lake sediments: a new method of endmember modeling using hierarchical clustering. *Quat. Res.* 89, 365–373. <https://doi.org/10.1017/qua.2017.78>
- Zhu, F., Emile-Geay, J., McKay, N.P., Hakim, G.J., Khider, D., Ault, T.R., Steig, E.J., Dee, S., Kirchner, J.W., 2019. Climate models can correctly simulate the continuum of global-average temperature variability. *Proc. Natl. Acad. Sci.* 116, 8728–8733. <https://doi.org/10.1073/pnas.1809959116>
- Zielhofer, C., von Suchodoletz, H., Fletcher, W.J., Schneider, B., Dietze, E., Schlegel, M., Schepanski, K., Weninger, B., Mischke, S., Mikdad, A., 2017. Millennial-scale fluctuations in Saharan dust supply across the decline of the African Humid Period. *Quat. Sci. Rev.* 171, 119–135. <https://doi.org/10.1016/j.quascirev.2017.07.010>



1199 12. Figures

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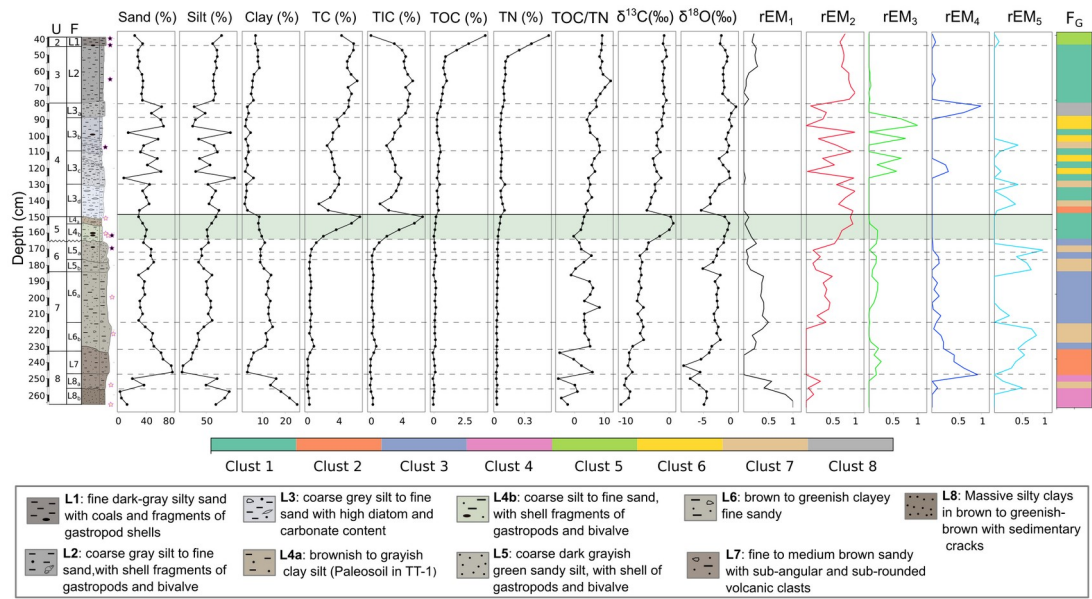
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**Figure 1.** Climate Dynamics and Stratigraphic Correlations at the ATTL. (A) Correlation between monthly-mean 500 hPa zonal-wind anomalies and precipitation in the ATTL region of central Chile, calculated from ERA-5 reanalysis data. The map shows the principal low-level atmospheric flows over South America, including the southern westerly winds (SWW), the southeast Pacific anticyclone (SPA), trade winds (TW), the Intertropical Convergence Zone (ITCZ), the subtropical low-level jet (LLJ), the South Atlantic Convergence Zone (SACZ), and the Antarctic Circumpolar Current (ACC), modified from Garreaud et al. (2009). Marine core locations along the Pacific coast are also indicated: 1 – SO147-106KL (12°S), 2 – GeoB3375/15016 (27°S), 3 – GeoB7139-2 (30°S), and 4 – ODP1233/GeoB7196-1 (41°S). (B) Hydrological basin, topography, and locations of Taguatagua archaeological sites —TT-1 (Montané, 1968; Varela, 1976), TT-2 (Nuñez et al., 1994), and TT-3 (Labarca et al., 2024) — and nearby Taguatagua paleoclimatic records (red star: Heusser, 1990; Valero-



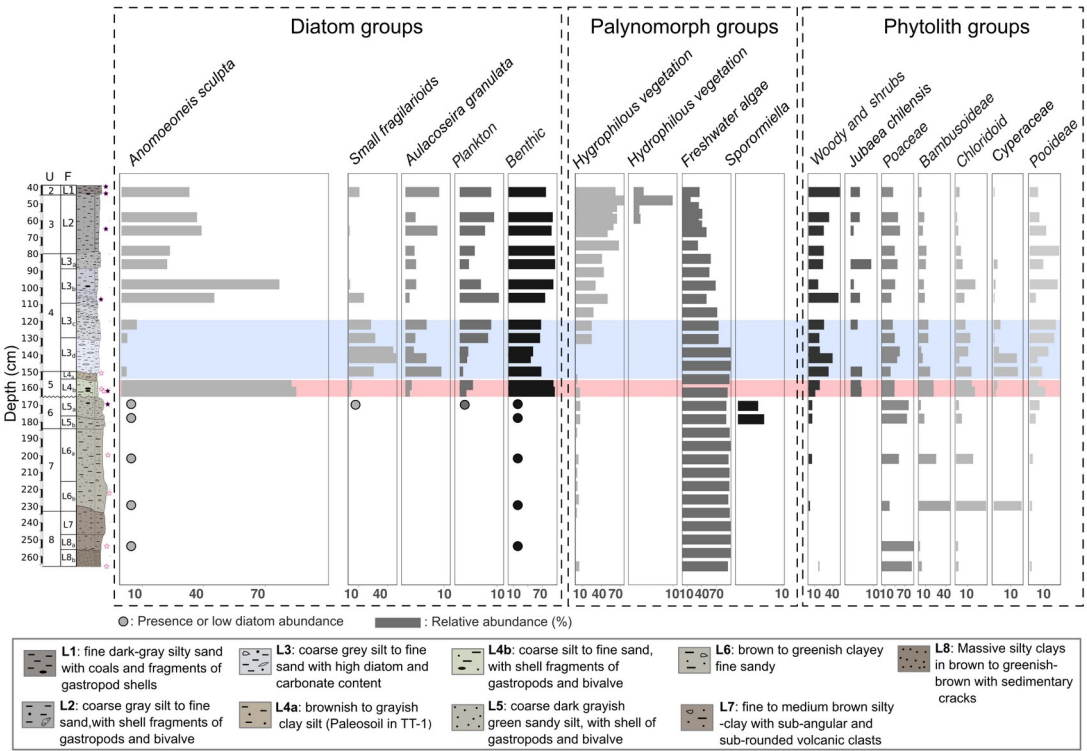
1214           Garcés et al., 2005) and (C) Stratigraphic correlation of geoarchaeological and  
1215           paleoclimatic profiles from the ATTL area.

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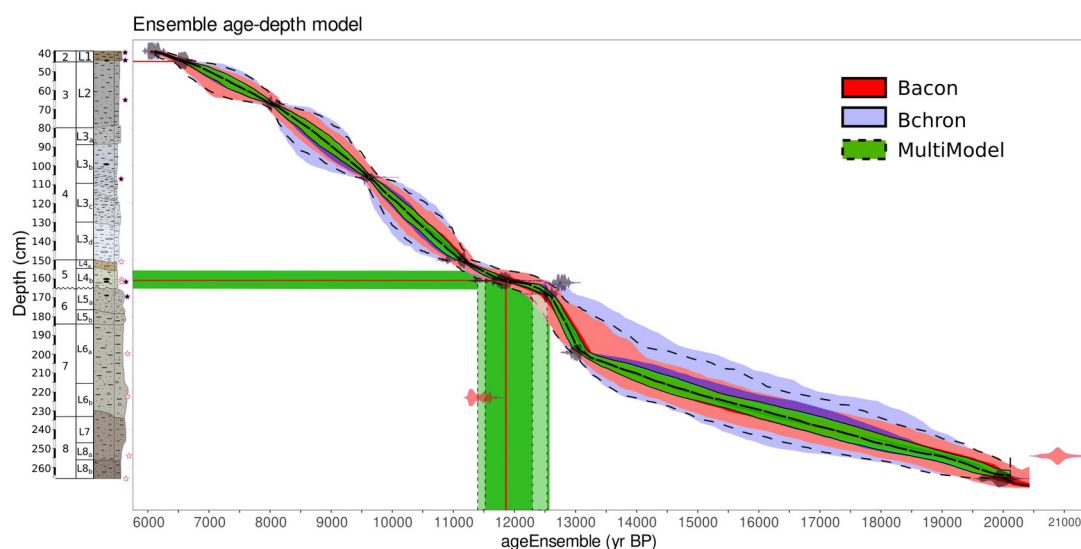
1217 **Figure 2.** Sedimentary Facies (F) and Units (U) of the TT19-3A Sequence. Stratigraphic  
1218 profile showing grain-size distributions (sand, silt, clay) and geochemical proxies—total  
1219 carbon (TC), total inorganic carbon (TIC), total organic carbon (TOC), total nitrogen (TN),  
1220 TOC/TN ratio, and the isotopic compositions of  $\delta^{13}\text{C}$  and  $\delta^{18}\text{O}$ . The figure also presents end-  
1221 member models (EM1–EM5) and cluster classifications (Clusters 1–8), which represent  
1222 distinct granulometric and geochemical associations. The shaded green region indicates the  
1223 onset of the Holocene.

1224

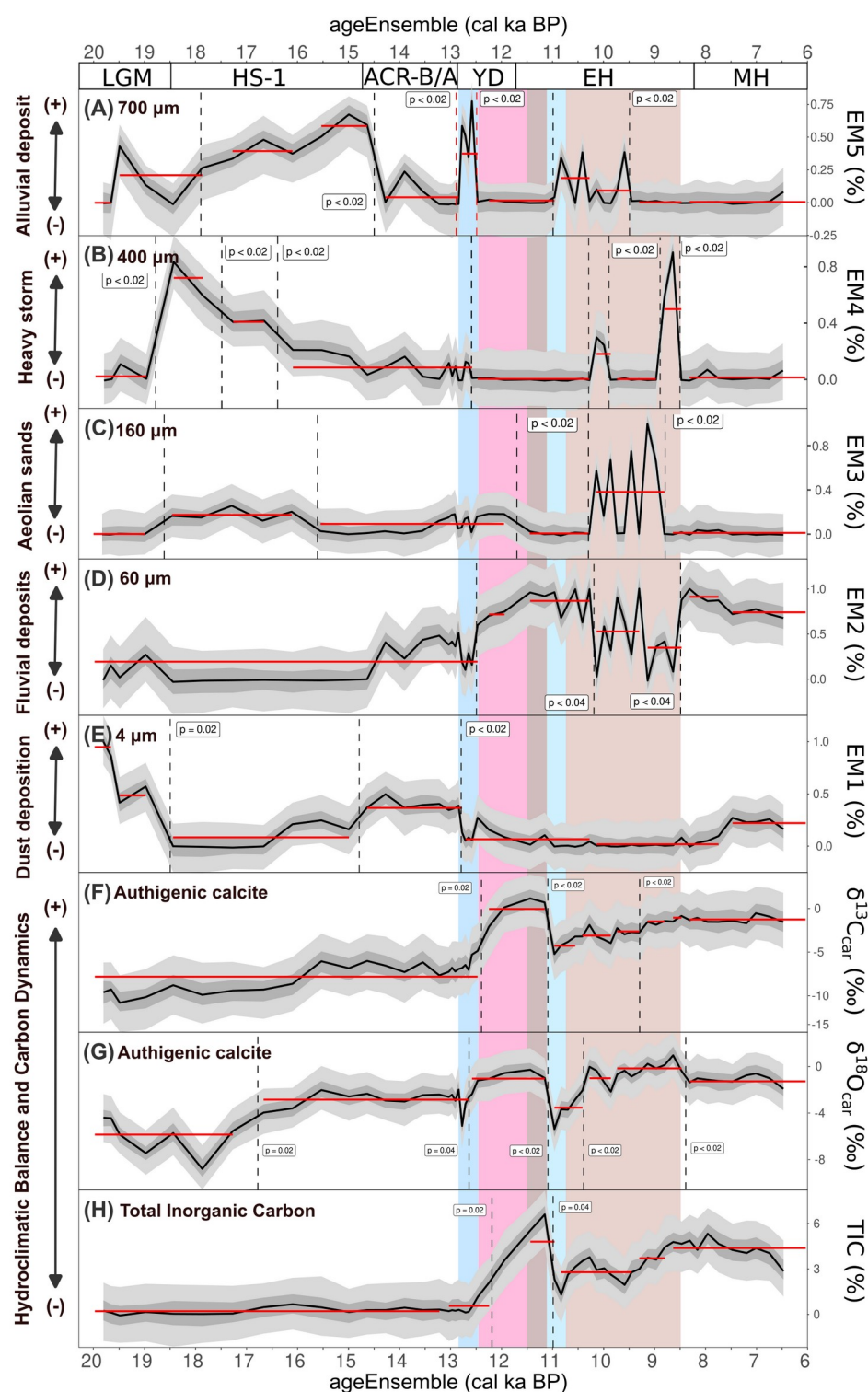


1225 **Figure 3. Biostratigraphy of Aquatic–Terrestrial proxies in the TT19-3A sequence.** The  
1226 figure shows changes in relative abundances of diatom, palynomorph, and phytolith  
1227 functional groups throughout the sequence. Diatoms are categorized as small fragilarioids,  
1228 *Aulacoseira granulata*, planktonic, and benthic types. Palynomorphs include hygrophytes,  
1229 hydrophytes, freshwater algae, and *Sporormiella*. Phytoliths comprise woody and shrub  
1230 types, *Jubaea chilensis*, Poaceae, Bambusoideae, Chloridoid, Cyperaceae, and Pooideae.  
1231 Sedimentary facies and unit descriptions are shown on the left. Blue and red shaded areas  
1232 indicate significant lake level changes during the recorded period.

1233



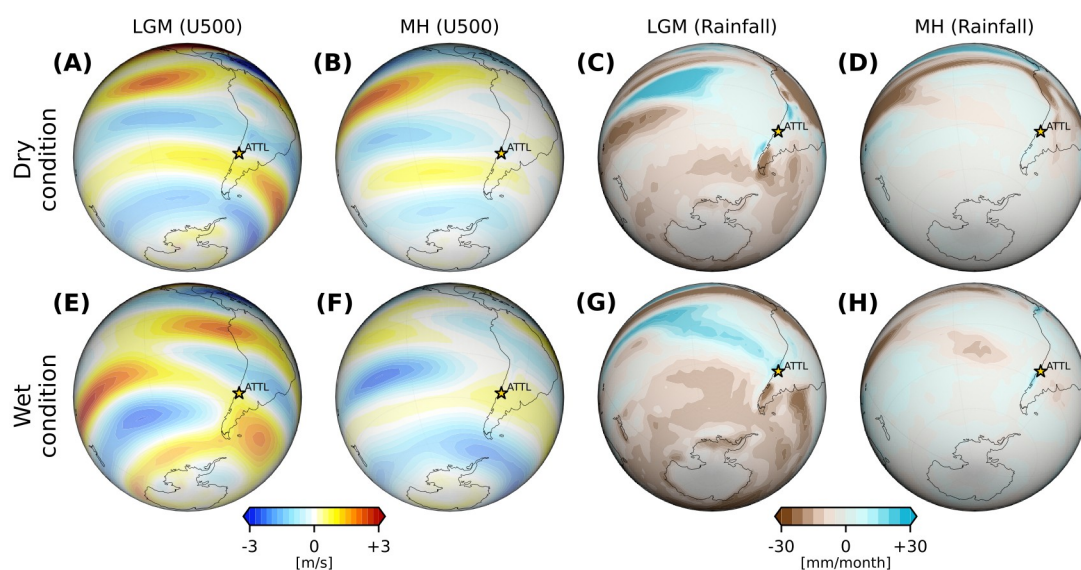
1234 **Figure 4.** Ensemble age-depth model for core TT19-3A. The figure shows age modeling  
 1235 results for core TT19-3A using two Bayesian methods: Bacon (red) and Bchron (blue). The  
 1236 green shading represents the combined multimodel output, with shaded areas indicating  
 1237 95% confidence intervals. Shaded green region marking the onset of the Holocene.



1239 **Figure 5.** Multi-proxy record from core TT19-3A showing grain size distributions, elemental  
 1240 and isotopic data over the past 20.0 ka BP. Showing key climatic intervals: Last Glacial  
 1241 Maximum (LGM), Heinrich Stadial 1 (HS-1), Antarctic Cold Reversal/Bølling-Allerød (ACR-  
 1242 BA), Younger Dryas (YD), Early Holocene (EH), and Middle Holocene (MH). Panels (a) to  
 1243 (d) display particle size distributions for different grain size fractions: (A) 700  $\mu\text{m}$  (alluvial  
 1244 deposits), (B) 400  $\mu\text{m}$  (heavy storm layers), (C) 160  $\mu\text{m}$  (aeolian sands), and (D) 60  $\mu\text{m}$   
 1245 (fluvial deposits). Panel (E) represents fine dust particles ( $<4 \mu\text{m}$ ). Panels (F) to (H) show

1246 geochemical proxies for authigenic calcite, including  $\delta^{13}\text{C}$  (per mil),  $\delta^{18}\text{O}$  (per mil), and total  
1247 inorganic carbon (TIC). Shaded areas highlight climatic transitions, and the black lines  
1248 represent ensemble averages with confidence intervals (grey shading). Red and black lines  
1249 indicate statistically significant changes in the time series ( $p < 0.02$  and  $p < 0.04$ ,  
1250 respectively).

1251



1252 **Figure 6.** Zonal wind and precipitation anomalies from the PMIP4 multi-model  
 1253 ensemble for the LGM and Middle Holocene (MH) periods. Panels (A-D) show dry  
 1254 anomalies relative to the pre-Industrial simulation. Panels (A-B) depict zonal wind  
 1255 (m/s), while panels (C-D) illustrate precipitation (mm/month). Figures (E-H) show wet  
 1256 anomalies for the same periods, with panels (E-F) showing zonal wind (m/s) and  
 1257 panels (G-H) displaying precipitation (mm/month). The yellow star marks the location  
 1258 of the ALTTL.



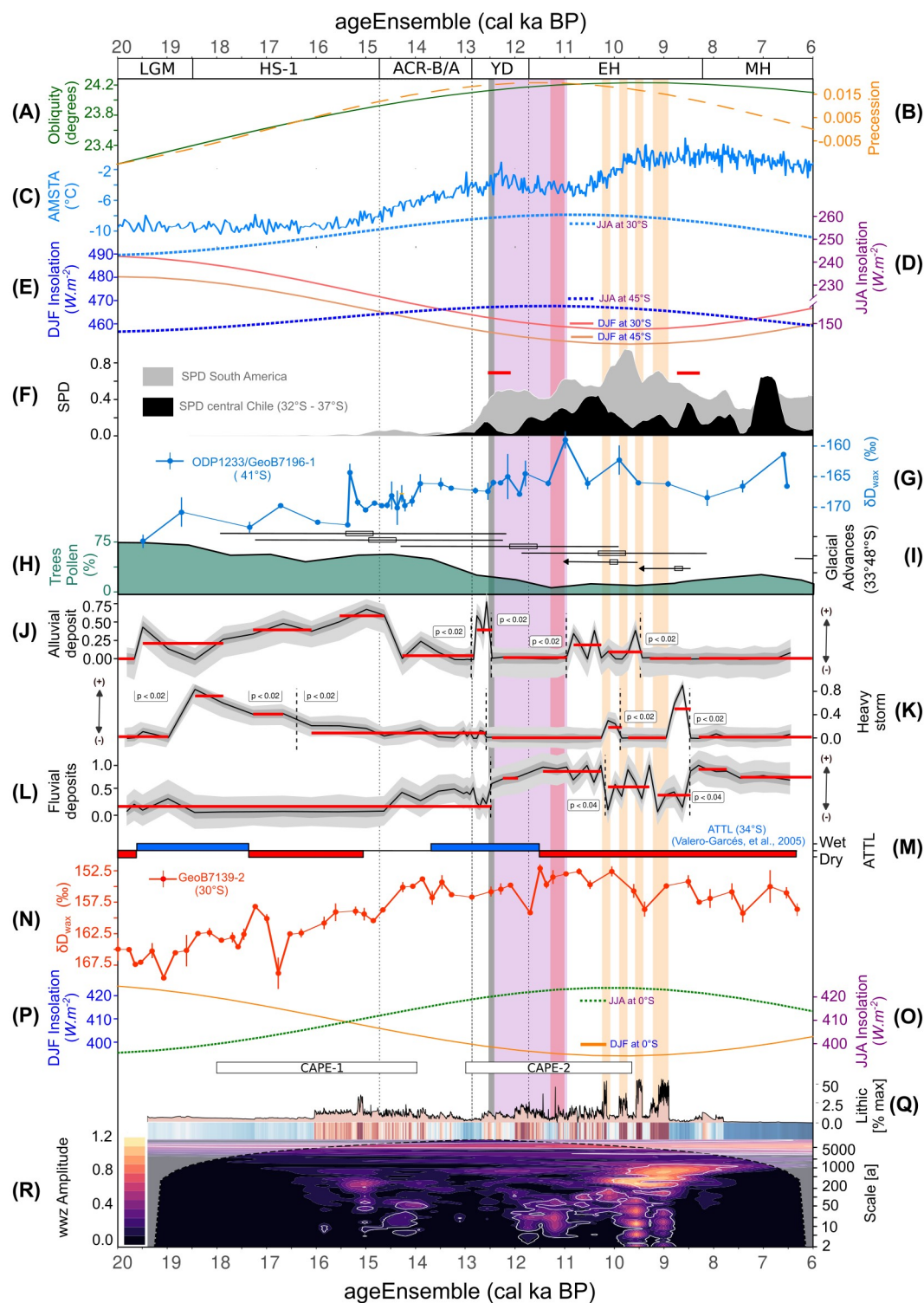


Figure 7. Multi-proxy comparison of climatic and environmental indicators for the ALTT over the last 20 cal ka BP. (A) Obliquity (green line) and (B) Precession (orange line) cycles. (C) Antarctic temperature anomaly (AMSG7TA, °C) from EPICA Dome C. (D) Winter (DJF) and (E) Summer (JJA) insolation curves ( $W \cdot m^{-2}$ ) at 30°S and 45°S. (F) Summed Probability Distributions (SPD) of archaeological dates: central Chile (32°S–37°S, black shading) and South America (grey shading). (G)  $\delta D_{wax}$  values from marine core ODP1233/GeoB7196-1 (41°S). (H) Tree pollen percentage from the ATTL. (I) Evidence of Andean glacial advances



1267 (33°–34°S). (J–L) Sedimentary features in the ATTL: relative frequency of alluvial deposits  
1268 (J), heavy storms (K), and fluvial deposits (L). (M) Wet and dry periods in the ATTL based on  
1269 Valero-Garcés et al., 2005. (N)  $\delta D_{wax}$  values from GeoB7139-2, indicating changes in  
1270 moisture sources and aridity. (O) JJA and (P) DJF insolation for 0°S. (Q) Sedimentary lithic  
1271 content (% max) and (R) Wavelet analysis (WWZ amplitude) from the 106KL marine  
1272 sedimentary record off the Peruvian coast. CAPE: Central Andes Precipitation Events.  
1273 Shaded vertical bands highlight major climatic periods and events: LGM (Last Glacial  
1274 Maximum), HS-1 (Heinrich Stadial 1), ACR-B/A (Antarctic Cold Reversal–Bølling/Allerød),  
1275 YD (Younger Dryas), EH (Early Holocene), and MH (Middle Holocene).